AITasker MVP — 18 Main Flows (Scope-Reduced · 28 Tables)

Cross-Table CRUD Grounding · State Machines · Endpoint Mapping

> **Purpose:** Definitive flow reference for teacher assessment. Every step is grounded to a specific table, column, state transition, and API endpoint from the 28-table MVP schema.
> **Last updated:** June 2026
> **Conventions:** `[LEDGER]` = wallet_transactions row written. `[API]` = external service call. Tables in **bold** on first reference. §0.x = Master Reference Sheet section.

Group 1 — Onboarding & Account Setup

MF-1: Client (CEO) Registration & Subscription

Overview

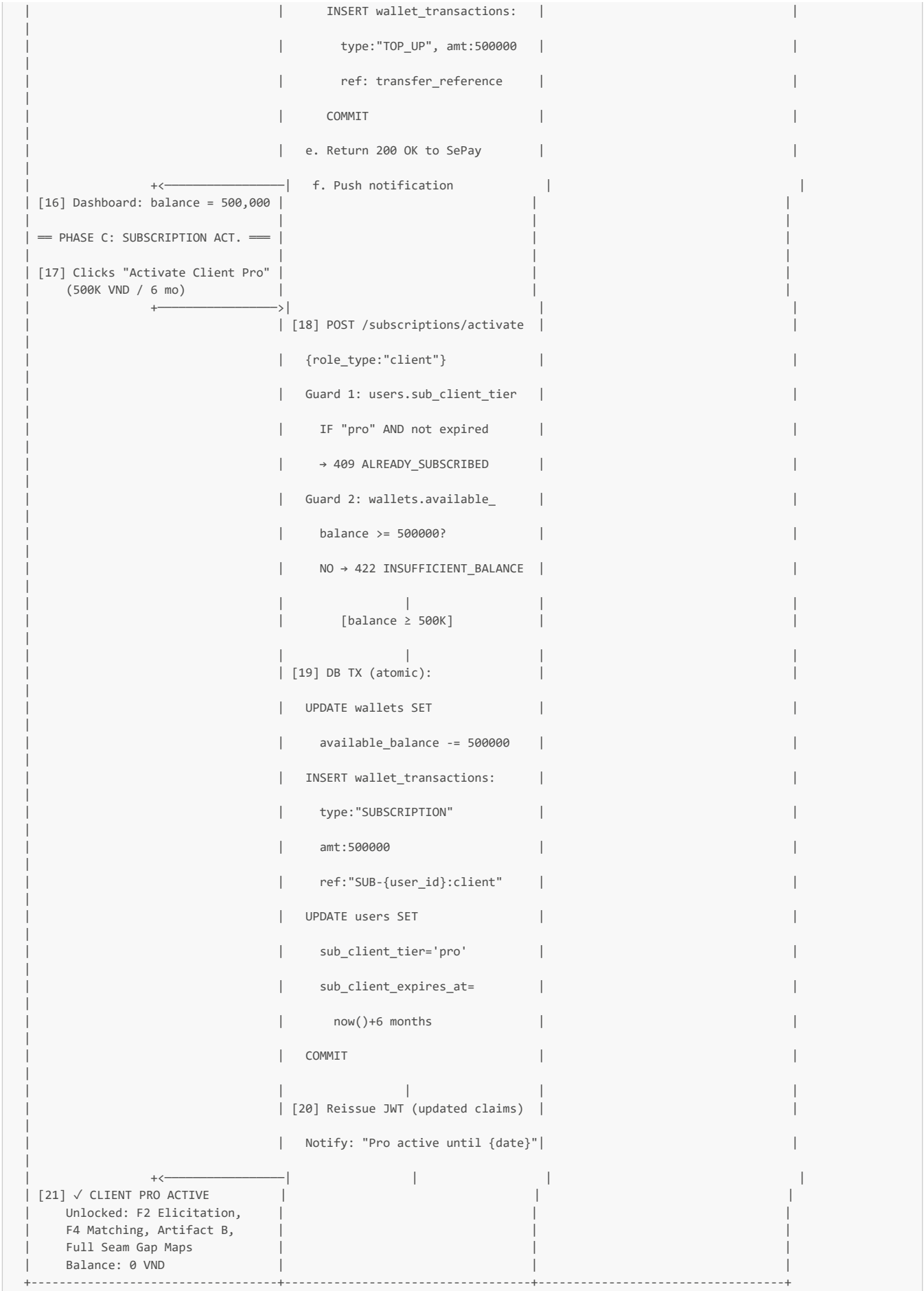
Registers a CLIENT/CEO account, creates their internal wallet and permanent top-up VA, then activates Client Pro subscription via direct wallet deduction. References: §0.7 (RBAC), §0.8 (Payment Architecture), §0.9 (Feature Gates).

Tables touched (5): `users`, `client_profiles`, `wallets`, `virtual_accounts`, `wallet_transactions`

Endpoints: `POST /auth/register`, `POST /wallet/topup-qr`, `POST /subscriptions/activate`, `POST /webhooks/sepay/ipn`

ASCII Swimlane


[5] POST SePay: Create VA
entity_type=WALLET_TOPUP
[6] SePay creates VA
Returns {va_number}
[7] INSERT virtual_accounts:
entity_type:"WALLET_TOPUP"
entity_id: user_id
va_number: (from SePay)
fixed_amount: NULL
expires_at: NULL (permanent)
status:"ACTIVE"
[8] Generate JWT:
{sub, active_role:"CLIENT",
client_subtype:"CEO",
roles:["CLIENT_CEO"]}
[9] CEO Dashboard (free tier)
Banner: "Subscribe to Pro"
All AI routes → 403
SUBSCRIPTION_REQUIRED
====== PHASE B: WALLET TOP-UP ===
[10] Clicks "Top Up Wallet"
[11] GET /wallet/topup-qr
SELECT virtual_accounts
WHERE entity_id=uid
AND entity_type='WALLET_TOPUP'
Generate VietQR from va_number
Return {qr_url, va_number}
[12] Scans VietQR, transfers
500,000 VND via banking app
[13] Bank processes transfer
[14] SePay IPN fires:
POST /webhooks/sepay/ipn
{va_number, amount, ref}
[15] IPN Handler:
a. Verify HMAC signature
b. SELECT virtual_accounts
WHERE va_number=?
→ entity_type=WALLET_TOPUP
→ entity_id=user_id
c. Idempotency: SELECT FROM
wallet_transactions WHERE
reference_id=? (UNIQUE idx)
d. DB TX (atomic):
UPDATE wallets SET
available_balance += 500000



Step Detail Table

Step	Actor	Action	Tables (CRUD)	State Change	Endpoint
1-2	CEO	Fill registration form	—	—	`GET /register`
3	NestJS	Validate input (email unique, pwd≥8, phone)	`users` (R)	—	`POST /auth/register`
4	NestJS	Atomic insert: user + client_profile + wallet	`users` (C), `client_profiles` (C), `wallets` (C)	New: user.roles=["CLIENT_CEO"], wallet available=0	—
5-6	NestJS→SePay	Create WALLET_TOPUP VA	— (SePay-side)	—	`POST SePay VA API`
7	NestJS	Store VA record	`virtual_accounts` (C)	VA status=ACTIVE	—
8	NestJS	Generate JWT	—	—	—
9	CEO	Dashboard loads free tier	`users` (R), `wallets` (R)	—	`GET /dashboard`
10-11	CEO→NestJS	Request top-up QR	`virtual_accounts` (R)	—	`POST /wallet/topup-qr`
12-14	CEO→Bank→SePay	External bank transfer + IPN	—	—	External
15	NestJS	IPN: verify HMAC → resolve VA → idempotency → credit wallet	`virtual_accounts` (R), `wallets` (U), `wallet_transactions` (C)	wallet.available: 0→500K	`POST /webhooks/sepay/ipn`
16	CEO	Dashboard reflects new balance	`wallets` (R)	—	`GET /wallet`
17-18	CEO→NestJS	Activate subscription; guards pass	`users` (R), `wallets` (R)	—	`POST /subscriptions/activate`
19	NestJS	Atomic: debit wallet + ledger + update user tier	`wallets` (U), `wallet_transactions` (C), `users` (U)	wallet.available: 500K→0, users.sub_client_tier: free→pro	—
20-21	NestJS→CEO	Reissue JWT, notify	`users` (R)	—	—

Cross-Table CRUD Map

```
users ──(1:1)──> client_profiles     [C in step 4]
users ──(1:1)──> wallets             [C in step 4, R/U in steps 15,19]
users               .subscription_client_tier  [R step 18, U step 19]
users               .sub_client_expires_at     [U step 19]
wallets ──(1:N)──> wallet_transactions [C in steps 15,19]
virtual_accounts    .entity_id → users.id (polymorphic) [C step 7, R steps 11,15]
wallet_transactions .reference_id UNIQUE index  [idempotency in step 15]
```

State Machine Reference

- **Subscription state (§0.9):** `users.subscription_client_tier`: `free` → `pro` (step 19); stored directly on `users` — no `user_subscriptions` table in MVP
- **VA state (§0.8):** `virtual_accounts.status`: `ACTIVE` (permanent for WALLET_TOPUP)
- **Wallet balance invariants:** `available_balance >= 0` (CHECK constraint), `locked_balance >= 0` (CHECK constraint)

MF-2: Expert Registration, Profile & Tier 1→2 Verification

Overview

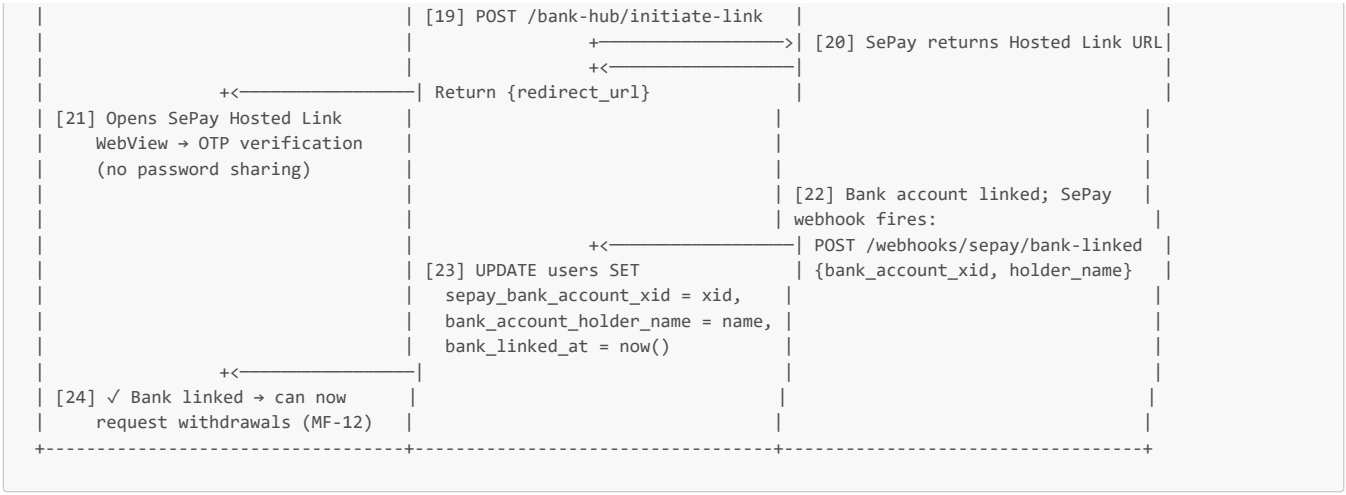
Registers an EXPERT account, creates taxonomy profile with domain depths and seam claims (Tier 1 Claimed), then optionally submits portfolio evidence for LLM auto-verification to Tier 2 (Evidence-backed). Links bank account via Bank Hub Hosted Link for future chi hộ withdrawals. References: §0.1 (Domains), §0.2 (Seams), §0.4 (2-Tier Verification).

Tables touched (8): `users`, `expert_profiles`, `wallets`, `virtual_accounts`, `wallet_transactions`, `expert_domain_depths`, `expert_seam_claims`, `portfolio_submissions`, `platform_decisions`

Endpoints: `POST /auth/register`, `PUT /expert-profile`, `POST /expert-profile/domains`, `POST /expert-profile/seams`, `POST /portfolio-submissions`, `GET /portfolio-submissions/{id}/eval-result`, `POST /bank-hub/initiate-link`, `POST /webhooks/sepay/bank-linked`

ASCII Swimlane


to Tier 2"
Guard: Expert Pro required
[12] Guard: users.sub_expert_tier
IF "free" → 403 SUBSCRIPTION_
REQUIRED
IF "pro" → proceed
[13] Fills portfolio form for
seam A↔C:
• project_description
• decision_points (structured)
[14] POST /portfolio-submissions
Throttle check:
SELECT submission_count,
locked_until FROM
expert_seam_claims
WHERE expert_id=? AND
seam_code='A↔C'
IF submission_count≥5 AND
locked_until>now()
→ 429 TOO_MANY_ATTEMPTS
INSERT portfolio_submissions:
expert_id, seam_claim_id,
project_description,
decision_points,
status:'PENDING',
submitted_at:now()
UPDATE expert_seam_claims SET
submission_count += 1
[15] FastAPI LLM extraction call
→ POST /llm/portfolio-eval
{project_description,
decision_points, seam_code}
FastAPI runs LLM rubric:
• All required signal types?
• Confidence score computed
• Returns {confidence, passed,
gap_advisory?}
[16] DB TX:
UPDATE portfolio_submissions SET
llm_confidence = {score},
evaluated_at = now(),
status = 'APPROVED'|'REJECTED'
IF confidence ≥ 0.85 AND passed:
UPDATE expert_seam_claims SET
verification_tier =
'EVIDENCE_BACKED'
(confidence factor: 0.55 per
§0.4)
INSERT platform_decisions:
decision_type:'SEAM_TIER_UPGRADE'
OR 'PORTFOLIO_EVAL',
entity_type:'expert_seam_claims',
entity_id:claim_id,
llm_confidence:{score},
decision:'APPROVED'|'REJECTED',
advisory_note:{gap_advisory}
IF rejected AND submission_count
≥ 5:
UPDATE expert_seam_claims SET
locked_until = now()+30days
[17] Result:
✓ A↔C → Tier 2 EVIDENCE_BACKED
✗ A↔D → remains Tier 1 CLAIMED
(gap advisory shown for
resubmission)
═══ PHASE D: BANK ACCOUNT LINK ═══
[18] Clicks "Link Bank Account"



Step Detail Table

Step	Actor	Action	Tables (CRUD)	State Change	Endpoint
1-3	Expert→NestJS	Register account	`users` (C), `expert_profiles` (C), `wallets` (C), `virtual_accounts` (C)	New expert user	`POST /auth/register`
5-6	Expert→NestJS	Set domain depths	`expert_domain_depths` (C/U via UPSERT)	verification_tier='CLAIMED' (Tier 1)	`POST /expert-profile/domains`
7-8	Expert→NestJS	Claim seams	`expert_seam_claims` (C)	verification_tier='CLAIMED'	`POST /expert-profile/seams`
9-10	Expert→NestJS	Set stack tags + engagement model + archetype history	`expert_profiles` (U)	—	`PUT /expert-profile`
11-12	Expert→NestJS	Tier 2 upgrade attempt; subscription guard	`users` (R)	—	`POST /portfolio-submissions`
13-14	Expert→NestJS	Submit portfolio evidence	`portfolio_submissions` (C), `expert_seam_claims` (U: submission_count++)	status='PENDING'	—
15	NestJS→FastAPI	LLM portfolio evaluation	—	—	`POST /llm/portfolio-eval` (internal)
16	FastAPI→NestJS	Process LLM result	`portfolio_submissions` (U), `expert_seam_claims` (U), `platform_decisions` (C)	seam verification_tier: CLAIMED→EVIDENCE_BACKED (if ≥0.85)	—
18-19	Expert→NestJS	Initiate Bank Hub link	—	—	`POST /bank-hub/initiate-link`
20-21	Expert→SePay	Complete OTP in Hosted Link	—	—	SePay WebView
22-23	SePay→NestJS	Bank linked webhook	`users` (U)	sepay_bank_account_xid set	`POST /webhooks/sepay/bank-linked`

Cross-Table CRUD Map

```
users ──(1:1)──> expert_profiles           [C step 3, U step 23]
expert_profiles ──(1:N)──> expert_domain_depths  [C step 6]
expert_profiles ──(1:N)──> expert_seam_claims    [C step 8, U steps 14,16]
expert_seam_claims ──(1:N)──> portfolio_submissions [C step 14, U step 16]
portfolio_submissions → platform_decisions  [C step 16 — polymorphic entity_id]
expert_profiles.stack_tags_json (JSONB)     [U step 10 — absorbs cut expert_stack_tags table]
expert_profiles.archetype_history_json (JSONB) [U step 10 — self-declared for cold start]
```

Key scope-reduction note: No `scenario_assessments`, `scenario_responses`, or `expert_seam_outcome_signals` tables. Expert verification maxes at Tier 2 (EVIDENCE_BACKED, confidence 0.55). Tiers 3-4 deferred to Phase 2.

MF-3: Tech Team Account Creation via Handoff Link

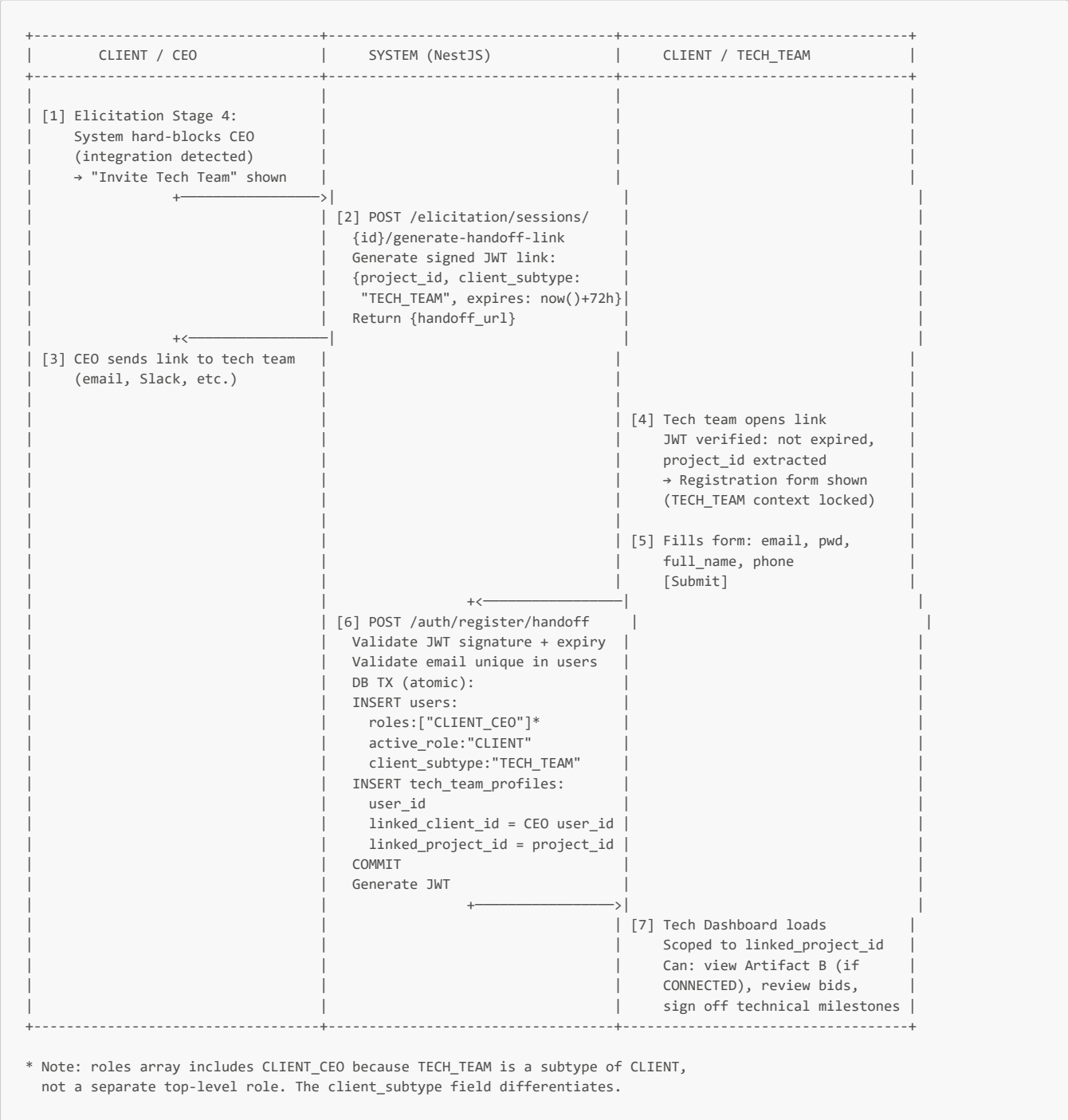
Overview

CEO generates a signed handoff link during Elicitation Stage 4. The link is sent to the client's technical team member, who uses it to register as CLIENT/TECH_TEAM. This is the **only** way a TECH_TEAM account is created — no self-registration. References: §0.7 (RBAC — TECH_TEAM creation constraint).

Tables touched (3): `users`, `tech_team_profiles`, `projects`

Endpoints: `POST /elicitation/sessions/{id}/generate-handoff-link`, `POST /auth/register/handoff`

ASCII Swimlane



* Note: roles array includes CLIENT_CEO because TECH_TEAM is a subtype of CLIENT, not a separate top-level role. The client_subtype field differentiates.

Step Detail Table

Step	Actor	Action	Tables (CRUD)	State Change	Endpoint

Step	Actor	Action	Tables (CRUD)	State Change	Endpoint
1-2	CEO→NestJS	Generate handoff link (signed JWT with project_id + 72h expiry)	`projects` (R — validate project exists)	—	`POST /elicitation/sessions/{id}/generate-handoff-link`
3	CEO	Sends link externally	—	—	—
4-5	TECH_TEAM	Opens link, fills form	—	—	`GET /register?token={jwt}`
6	NestJS	Validate handoff JWT + register TECH_TEAM	`users` (C), `tech_team_profiles` (C)	New TECH_TEAM user, scoped to one project	`POST /auth/register/handoff`
7	TECH_TEAM	Tech Dashboard loads	`tech_team_profiles` (R — get linked_project_id)	—	`GET /tech-dashboard`

Cross-Table CRUD Map

```
users ──(1:1)──> tech_team_profiles       [C step 6]
tech_team_profiles.linked_client_id → users.id (CEO) [C step 6]
tech_team_profiles.linked_project_id → projects.id   [C step 6 — scope lock]
tech_team_profiles linked_project_id INDEX  [R step 7 — fast project scoping]
```

Key constraint: TECH_TEAM account is **locked to one project** via `linked_project_id`. No self-registration. This enforces the §0.7 RBAC rule that only CEOs can initiate the tech team relationship.

Group 2 — Path A: Project-Based Flow

MF-4: AI Elicitation Engine (5-Stage)

Overview

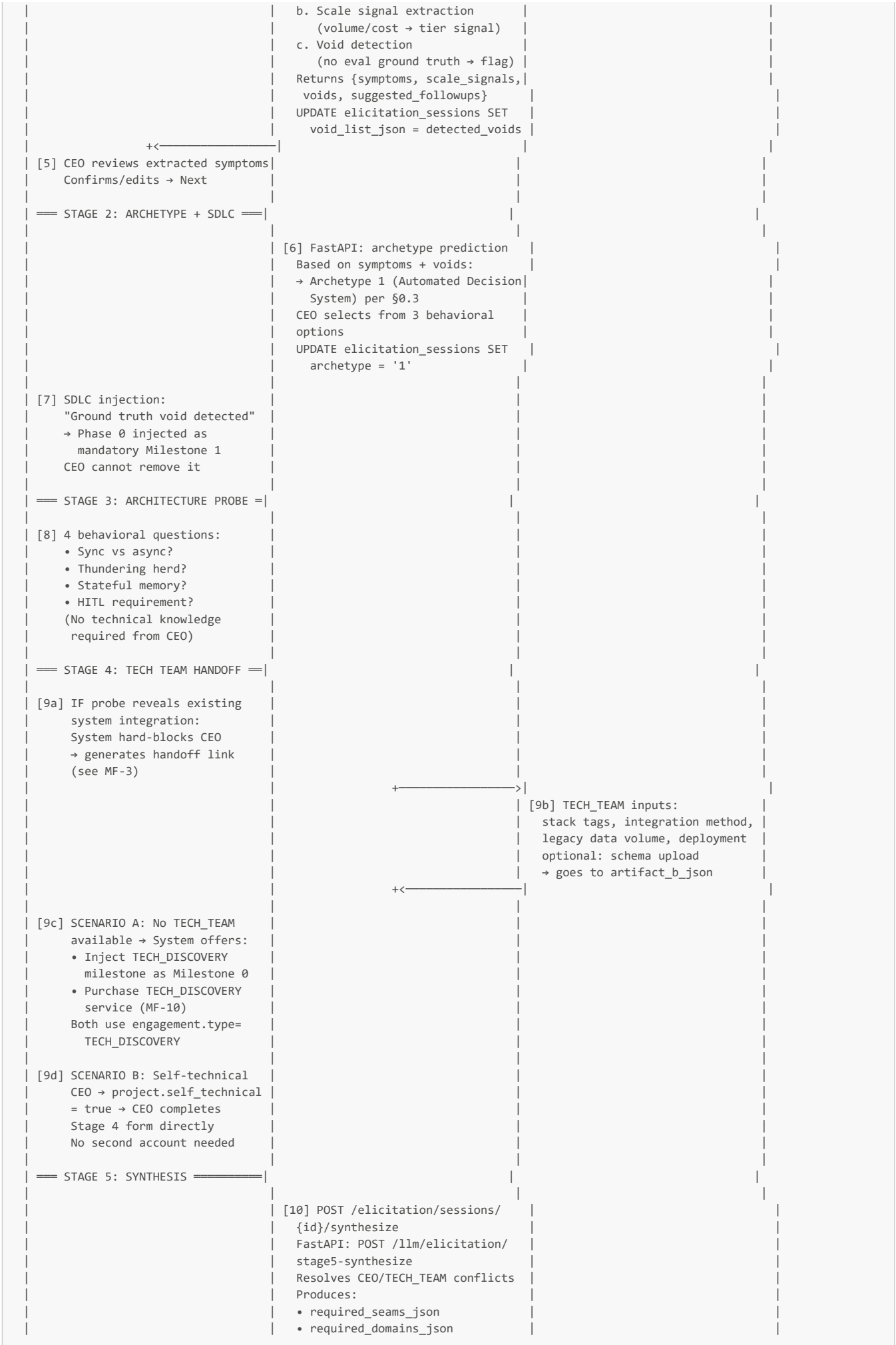
The platform's primary AI innovation. Transforms a CEO's raw symptom description into a taxonomy-grounded capability footprint (Artifact A + Artifact B) through a 5-stage conversational diagnostic. Supports Scenario A (no TECH_TEAM → TECH_DISCOVERY) and Scenario B (self-technical CEO). References: §0.1 (Domains), §0.2 (Seams), §0.3 (Archetypes + Tiers), §0.6 (Elicitation + Spec state machines).

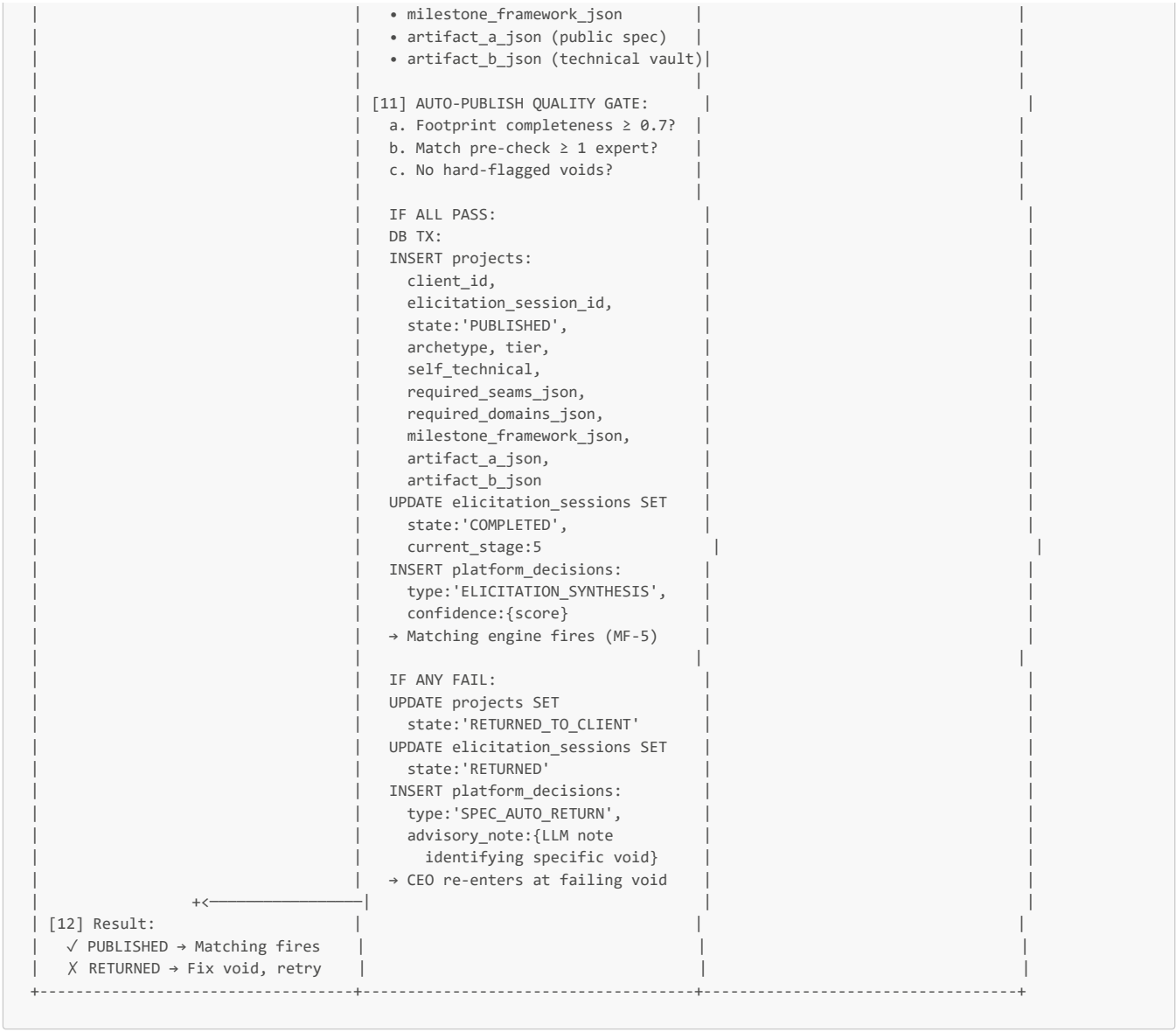
Tables touched (4): `elicitation_sessions`, `projects`, `platform_decisions`, `users`

Endpoints: `POST /elicitation/sessions`, `PUT /elicitation/sessions/{id}/stage`, `POST /elicitation/sessions/{id}/synthesize`, `POST /llm/elicitation/*` (FastAPI internal)

ASCII Swimlane

```
+-----------------------------------+-----------------------------------+-----------------------------------+
|        CLIENT / CEO               |   SYSTEM (NestJS + FastAPI)       |   CLIENT / TECH_TEAM              |
+-----------------------------------+-----------------------------------+-----------------------------------+
|                                   |                                   |                                   |
| ═══ STAGE 1: SYMPTOM INTAKE ═══════|                                  |                                   |
|                                   |                                   |                                   |
| [1] Clicks "Start New AI Project" |                                   |                                   |
|     Guard: sub_client_tier='pro'  |                                   |                                   |
|                +─────────────────>|                                   |                                   |
|                                   | [2] POST /elicitation/sessions    |                                   |
|                                   |   INSERT elicitation_sessions:    |                                   |
|                                   |     user_id, current_stage:1,     |                                   |
|                                   |     state:'IN_PROGRESS',          |                                   |
|                                   |     void_list_json:'[]'           |                                   |
|                                   |   Return session + stage 1 prompt |                                   |
|                +<─────────────────|                                   |                                   |
| [3] Types free-form pain:         |                                   |                                   |
|     "Our AI compliance output     |                                   |                                   |
|      is inconsistent"             |                                   |                                   |
|                +─────────────────>|                                   |                                   |
|                                   | [4] FastAPI: POST /llm/elicitation/|                                  |
|                                   |   stage1-extract                  |                                   |
|                                   |   LLM runs 3 extraction passes:   |                                   |
|                                   |   a. Intent separation            |                                   |
|                                   |      (strips "we want to fine-tune"|                                  |
|                                   |       → symptom "compliance       |                                   |
|                                   |       inconsistent")              |                                   |
```


b. Scale signal extraction
(volume/cost → tier signal)
c. Void detection
(no eval ground truth → flag)
Returns {symptoms, scale_signals,
voids, suggested_followups}
UPDATE elicitation_sessions SET
void_list_json = detected_voids
[5] CEO reviews extracted symptoms
Confirms/edits → Next
=== STAGE 2: ARCHETYPE + SDLC ===
[6] FastAPI: archetype prediction
Based on symptoms + voids:
→ Archetype 1 (Automated Decision
System) per §0.3
CEO selects from 3 behavioral
options
UPDATE elicitation_sessions SET
archetype = '1'
[7] SDLC injection:
"Ground truth void detected"
→ Phase 0 injected as
mandatory Milestone 1
CEO cannot remove it
=== STAGE 3: ARCHITECTURE PROBE =
[8] 4 behavioral questions:
• Sync vs async?
• Thundering herd?
• Stateful memory?
• HITL requirement?
(No technical knowledge
required from CEO)
=== STAGE 4: TECH TEAM HANDOFF ==
[9a] IF probe reveals existing
system integration:
System hard-blocks CEO
→ generates handoff link
(see MF-3)
[9b] TECH_TEAM inputs:
stack tags, integration method,
legacy data volume, deployment
optional: schema upload
→ goes to artifact_b_json
[9c] SCENARIO A: No TECH_TEAM
available → System offers:
• Inject TECH_DISCOVERY
milestone as Milestone 0
• Purchase TECH_DISCOVERY
service (MF-10)
Both use engagement.type=
TECH_DISCOVERY
[9d] SCENARIO B: Self-technical
CEO → project.self_technical
= true → CEO completes
Stage 4 form directly
No second account needed
=== STAGE 5: SYNTHESIS ==========
[10] POST /elicitation/sessions/
{id}/synthesize
FastAPI: POST /llm/elicitation/
stage5-synthesize
Resolves CEO/TECH_TEAM conflicts
Produces:
• required_seams_json
• required_domains_json



Step Detail Table

Step	Actor	Action	Tables (CRUD)	State Change	Endpoint
1-2	CEO→NestJS	Start elicitation session; subscription guard	`users` (R — guard), `elicitation_sessions` (C)	session.state=IN_PROGRESS, current_stage=1	`POST /elicitation/sessions`
3-4	CEO→FastAPI	Symptom intake; LLM extraction	`elicitation_sessions` (U — void_list_json)	—	`POST /llm/elicitation/stage1-extract`
5-6	CEO→FastAPI	Archetype confirmation + SDLC injection	`elicitation_sessions` (U — archetype)	—	`PUT /elicitation/sessions/{id}/stage`
7	System	Void injection (Phase 0 mandatory)	`elicitation_sessions` (U — void_list_json)	—	—
8	CEO→NestJS	Architecture probe (4 behavioral questions)	`elicitation_sessions` (U — current_stage=3)	—	`PUT /elicitation/sessions/{id}/stage`
9a-b	CEO/TECH_TEAM	Tech team handoff (see MF-3)	`users` (C), `tech_team_profiles` (C)	—	`POST /auth/register/handoff`
9c	CEO→NestJS	Scenario A: TECH_DISCOVERY offer	`elicitation_sessions` (U — scenario_type='SCENARIO_A')	—	—

Step	Actor	Action	Tables (CRUD)	State Change	Endpoint
9d	CEO→NestJS	Scenario B: self_technical=true	elicitation_sessions (U — scenario_type='SCENARIO_B')	—	PUT /elicitation/sessions/{id}/stage
10	NestJS→FastAPI	Stage 5 synthesis	—	—	POST /llm/elicitation/stage5-synthesize
11	NestJS	Auto-publish quality gate + project creation	projects (C), elicitation_sessions (U), platform_decisions (C)	projects.state: PUBLISHED or RETURNED_TO_CLIENT; session.state: COMPLETED or RETURNED	POST /elicitation/sessions/{id}/synthesize
12	CEO	View result	projects (R)	—	—

State Machine References

Elicitation session (§0.6):

```
IN_PROGRESS (current_stage advances 1→2→3→4→5)
  → COMPLETED (Stage 5 synthesis produced project; project.state=PUBLISHED)
  → RETURNED  (auto-publish gate failed; re-enter at current_stage)
  → ABANDONED (CEO navigated away; can resume)
```

Spec/Project (§0.6):

```
DRAFT → [auto-publish quality gate] → PUBLISHED
                    ↓ (fail)
            RETURNED_TO_CLIENT (LLM advisory note; re-enter at specific void)
PUBLISHED → SUSPENDED (admin emergency only)
```

Cross-table dependency:

```
elicitation_sessions.user_id → users.id
projects.elicitation_session_id → elicitation_sessions.id
projects.client_id → users.id
projects self-contains: required_seams_json, required_domains_json,
  milestone_framework_json, artifact_a_json, artifact_b_json
  (5 JSONB columns replace 3 cut tables: capability_footprints, artifact_a, artifact_b)
```

MF-5: AI Matching & Shortlisting (2-Tier Confidence)

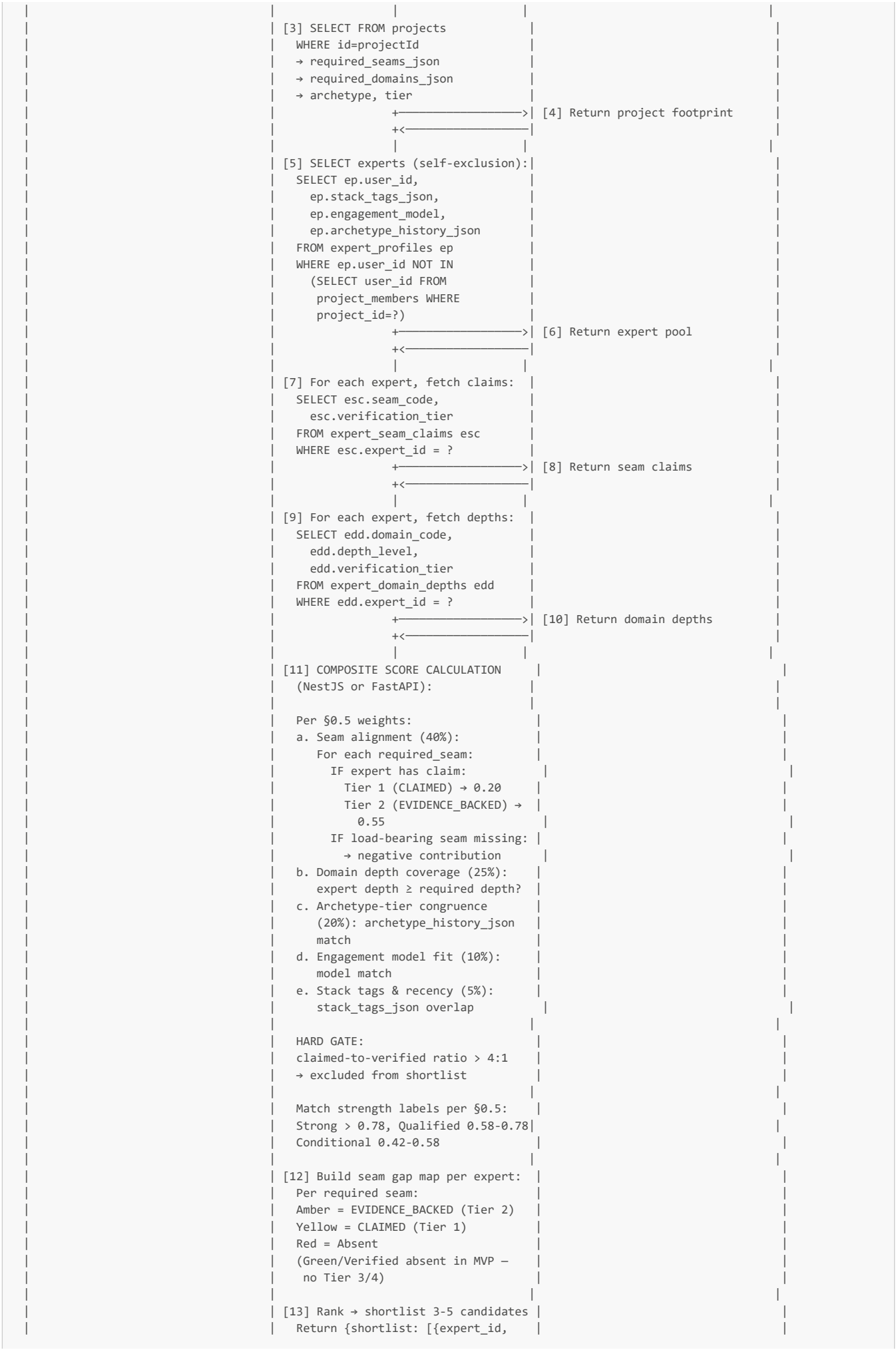
Overview

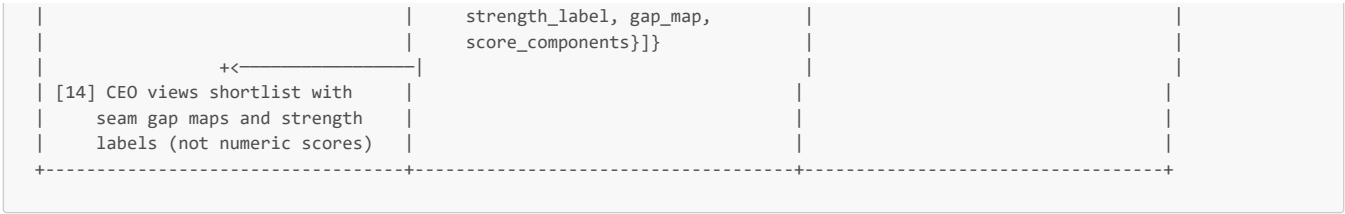
Composite scoring engine ranks experts against a project's capability footprint using the 5-component formula from §0.5, with only Tier 1 (0.20) and Tier 2 (0.55) confidence weights per §0.4. Produces a shortlist of 3-5 candidates with seam gap maps.

Tables touched (4): projects, expert_profiles, expert_seam_claims, expert_domain_depths

Endpoints: POST /matching/{projectId}, GET /matching/{projectId}/shortlist

ASCII Swimlane


[3] SELECT FROM projects
WHERE id=projectId
→ required_seams_json
→ required_domains_json
→ archetype, tier
[4] Return project footprint
[5] SELECT experts (self-exclusion):
SELECT ep.user_id,
ep.stack_tags_json,
ep.engagement_model,
ep.archetype_history_json
FROM expert_profiles ep
WHERE ep.user_id NOT IN
(SELECT user_id FROM
project_members WHERE
project_id=?)
[6] Return expert pool
[7] For each expert, fetch claims:
SELECT esc.seam_code,
esc.verification_tier
FROM expert_seam_claims esc
WHERE esc.expert_id = ?
[8] Return seam claims
[9] For each expert, fetch depths:
SELECT edd.domain_code,
edd.depth_level,
edd.verification_tier
FROM expert_domain_depths edd
WHERE edd.expert_id = ?
[10] Return domain depths
[11] COMPOSITE SCORE CALCULATION
(NestJS or FastAPI):
Per §0.5 weights:
a. Seam alignment (40%):
For each required_seam:
IF expert has claim:
Tier 1 (CLAIMED) → 0.20
Tier 2 (EVIDENCE_BACKED) →
0.55
IF load-bearing seam missing:
→ negative contribution
b. Domain depth coverage (25%):
expert depth ≥ required depth?
c. Archetype-tier congruence
(20%): archetype_history_json
match
d. Engagement model fit (10%):
model match
e. Stack tags & recency (5%):
stack_tags_json overlap
HARD GATE:
claimed-to-verified ratio > 4:1
→ excluded from shortlist
Match strength labels per §0.5:
Strong > 0.78, Qualified 0.58-0.78
Conditional 0.42-0.58
[12] Build seam gap map per expert:
Per required seam:
Amber = EVIDENCE_BACKED (Tier 2)
Yellow = CLAIMED (Tier 1)
Red = Absent
(Green/Verified absent in MVP —
no Tier 3/4)
[13] Rank → shortlist 3-5 candidates
Return {shortlist: [{expert_id,



Step Detail Table

Step	Actor	Action	Tables (CRUD)	Key Logic	Endpoint
1-2	CEO→NestJS	Initiate matching; guards	`users` (R), `projects` (R)	project must be PUBLISHED, user must be pro	`POST /matching/{projectId}`
3-4	NestJS	Read project footprint	`projects` (R)	Extract required_seams_json, required_domains_json, archetype, tier	—
5-6	NestJS	Query expert pool with self-exclusion	`expert_profiles` (R)	WHERE user_id NOT IN project members	—
7-10	NestJS	Fetch per-expert seam claims + domain depths	`expert_seam_claims` (R), `expert_domain_depths` (R)	verification_tier determines confidence factor (0.20 or 0.55)	—
11	NestJS/FastAPI	Compute composite scores	—	5-component formula per §0.5; hard gate ratio > 4:1	—
12-13	NestJS	Build gap maps + rank	—	Tier 2=Amber, Tier 1=Yellow, Absent=Red	—
14	CEO	View shortlist	—	—	`GET /matching/{projectId}/shortlist`

MF-6: Simplified Bid & Connection Flow

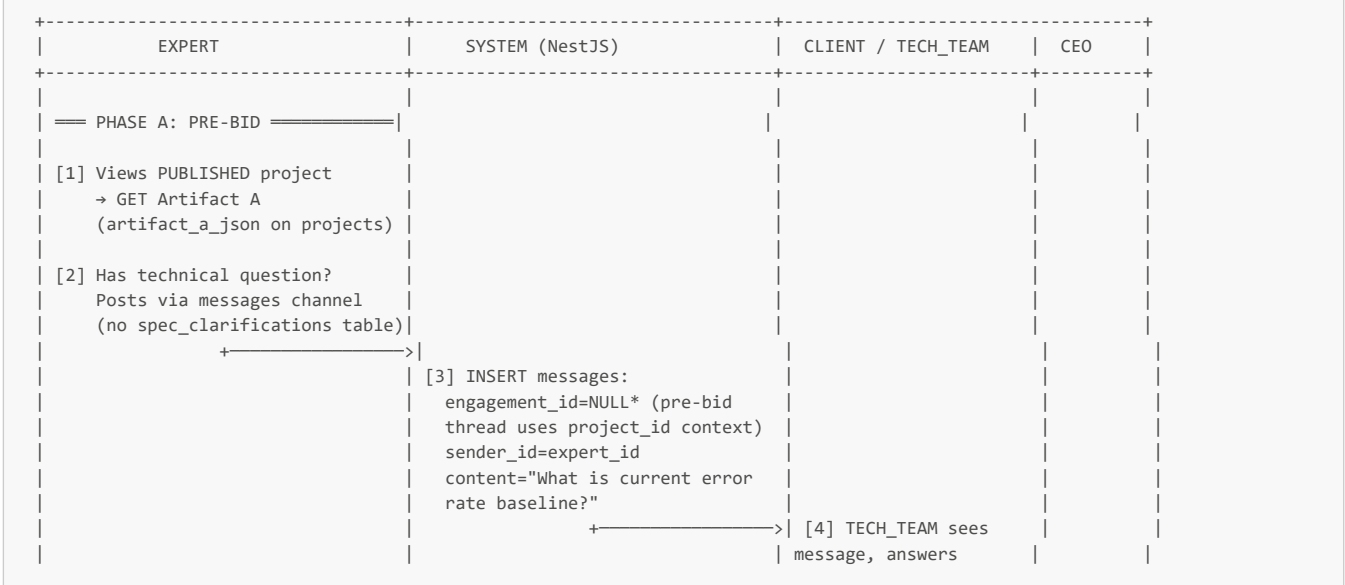
Overview

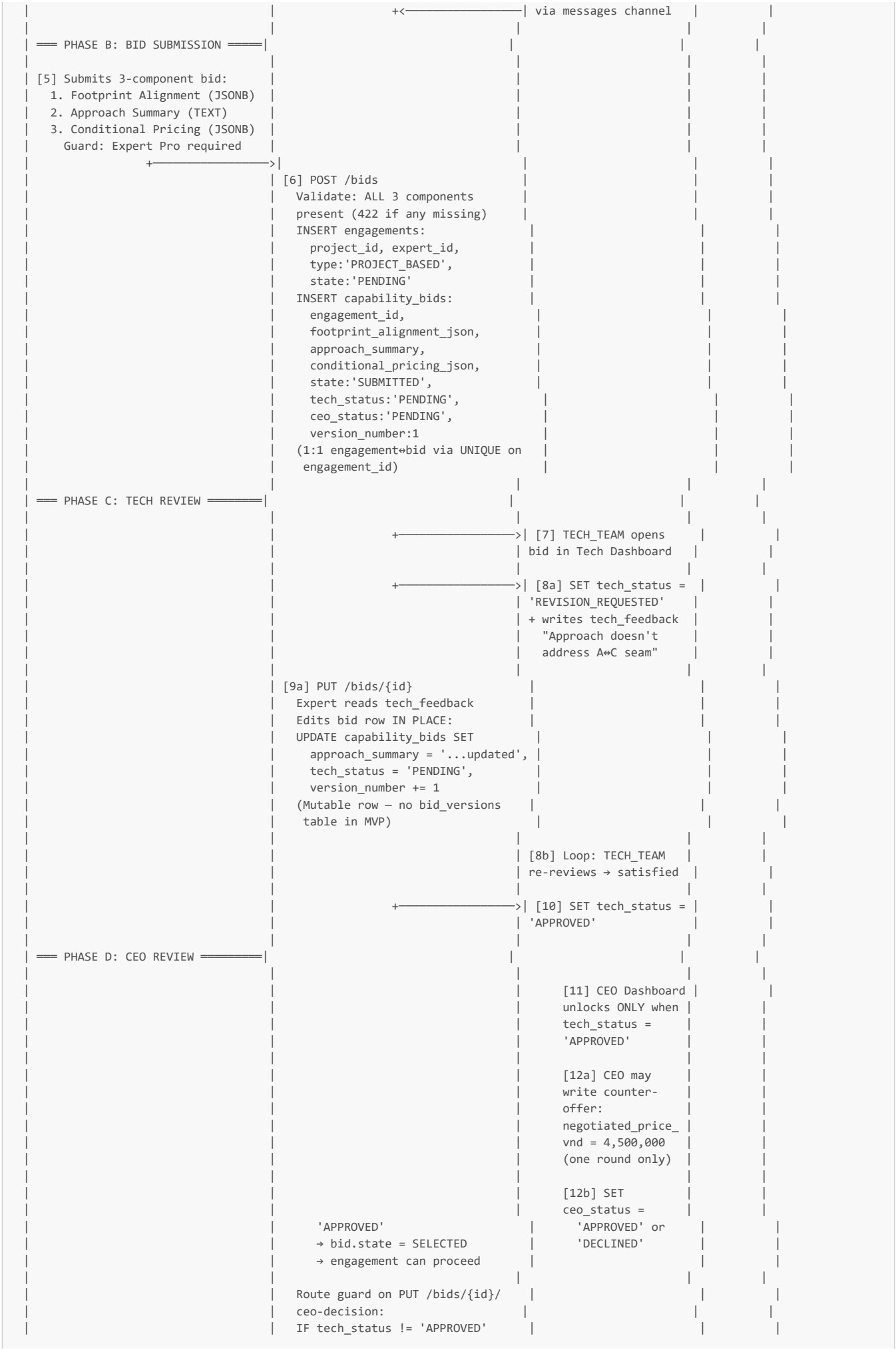
Expert views Artifact A, asks pre-bid questions via messages channel, submits a 3-component bid. TECH_TEAM reviews and sets tech_status. CEO reviews and sets ceo_status. Single mutable row — no versioned snapshots. Pre-bid questions use messages (no spec_clarifications table). Price negotiation via negotiated_price_vnd column (one round). References: §0.6 (Bid states), §0.7 (RBAC — TECH_TEAM approves before CEO).

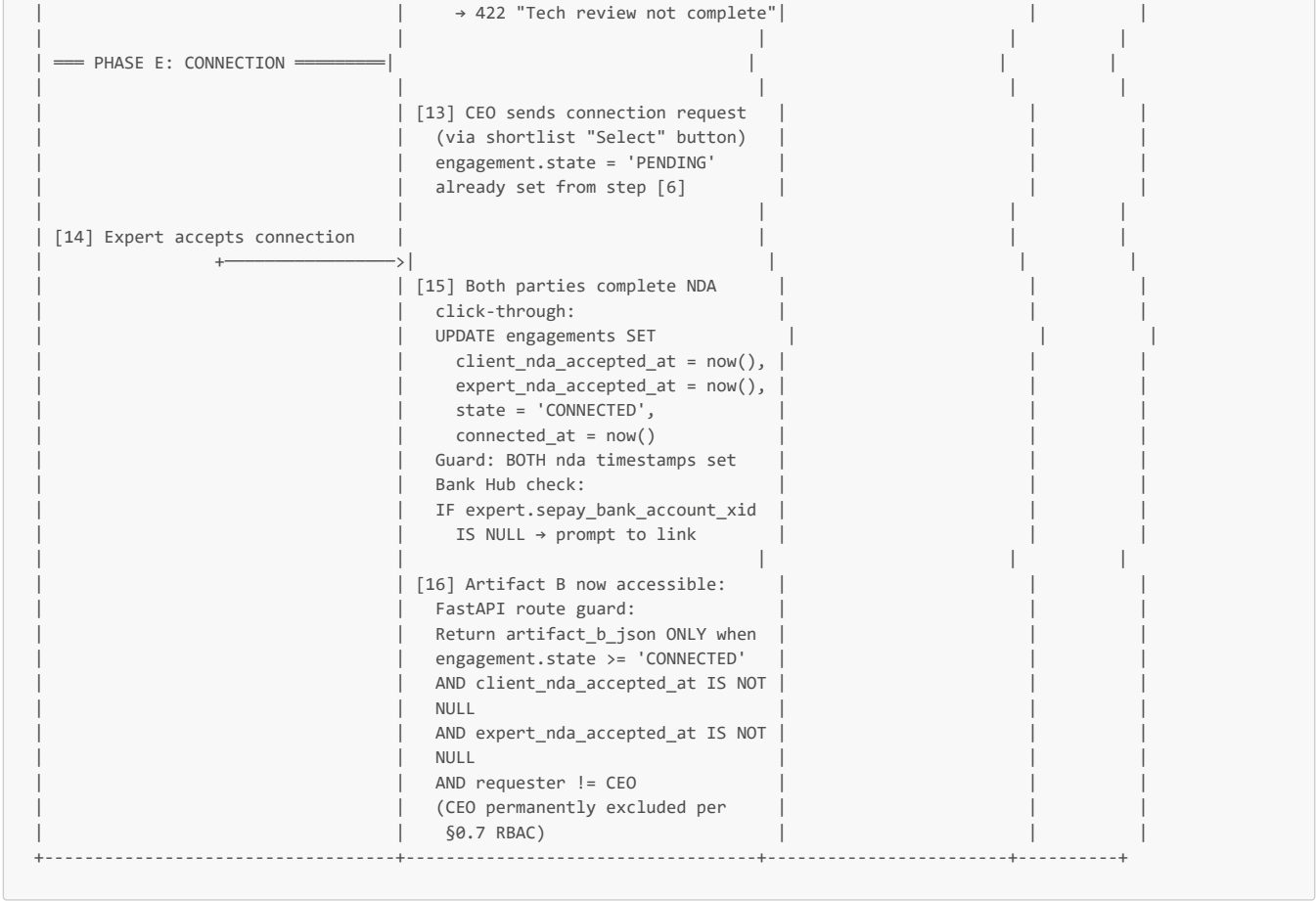
Tables touched (4): `capability_bids`, `engagements`, `projects`, `messages`

Endpoints: `GET /projects/{id}/artifact-a`, `POST /bids`, `PUT /bids/{id}`, `PUT /bids/{id}/tech-review`, `PUT /bids/{id}/ceo-decision`, `POST /engagements/{id}/connect`, `PUT /engagements/{id}/accept-nda`

ASCII Swimlane




via messages channel
=== PHASE B: BID SUBMISSION ===
[5] Submits 3-component bid:
1. Footprint Alignment (JSONB)
2. Approach Summary (TEXT)
3. Conditional Pricing (JSONB)
Guard: Expert Pro required
[6] POST /bids
Validate: ALL 3 components
present (422 if any missing)
INSERT engagements:
project_id, expert_id,
type:'PROJECT_BASED',
state:'PENDING'
INSERT capability_bids:
engagement_id,
footprint_alignment_json,
approach_summary,
conditional_pricing_json,
state:'SUBMITTED',
tech_status:'PENDING',
ceo_status:'PENDING',
version_number:1
(1:1 engagement↔bid via UNIQUE on
engagement_id)
=== PHASE C: TECH REVIEW ===
[7] TECH_TEAM opens
bid in Tech Dashboard
[8a] SET tech_status =
'REVISION_REQUESTED'
+ writes tech_feedback
"Approach doesn't
address A↔C seam"
[9a] PUT /bids/{id}
Expert reads tech_feedback
Edits bid row IN PLACE:
UPDATE capability_bids SET
approach_summary = '...updated',
tech_status = 'PENDING',
version_number += 1
(Mutable row — no bid_versions
table in MVP)
[8b] Loop: TECH_TEAM
re-reviews → satisfied
[10] SET tech_status =
'APPROVED'
=== PHASE D: CEO REVIEW ===
[11] CEO Dashboard
unlocks ONLY when
tech_status =
'APPROVED'
[12a] CEO may
write counter-
offer:
negotiated_price_
vnd = 4,500,000
(one round only)
[12b] SET
ceo_status =
'APPROVED'
→ bid.state = SELECTED
→ engagement can proceed
'APPROVED' or
'DECLINED'
Route guard on PUT /bids/{id}/
ceo-decision:
IF tech_status != 'APPROVED'



Step Detail Table

Step	Actor	Action	Tables (CRUD)	State Change	Endpoint
1	Expert	View Artifact A	`projects` (R — artifact_a_json)	—	`GET /projects/{id}/artifact-a`
2-3	Expert	Ask pre-bid question	`messages` (C)	—	`POST /engagements/{id}/messages`
4	TECH_TEAM	Answer via messages	`messages` (C)	—	`POST /engagements/{id}/messages`
5-6	Expert	Submit 3-component bid	`engagements` (C), `capability_bids` (C)	bid.state=SUBMITTED, tech_status=PENDING, ceo_status=PENDING	`POST /bids`
7-8a	TECH_TEAM	Set tech_status=REVISION_REQUESTED + tech_feedback	`capability_bids` (U)	tech_status: PENDING→REVISION_REQUESTED	`PUT /bids/{id}/tech-review`
9a	Expert	Edit bid row in place	`capability_bids` (U)	tech_status→PENDING, version_number++	`PUT /bids/{id}`
10	TECH_TEAM	Set tech_status=APPROVED	`capability_bids` (U)	tech_status: PENDING→APPROVED	`PUT /bids/{id}/tech-review`
11-12	CEO	Review + set ceo_status + optional negotiated_price_vnd	`capability_bids` (U)	ceo_status: PENDING→APPROVED/DECLINED	`PUT /bids/{id}/ceo-decision`
13	CEO	Connection request sent	`engagements` (R — already PENDING)	—	—
14-15	Expert	Accept + NDA click-through	`engagements` (U)	state: PENDING→CONNECTED; nda timestamps set	`POST /engagements/{id}/connect`, `PUT /engagements/{id}/accept-nda`
16	System	Artifact B route guard unlocks	`projects` (R — artifact_b_json)	—	`GET /projects/{id}/artifact-b`

Bid State Machine (Simplified — §0.6)

```
DRAFT → SUBMITTED → TECH_REVIEW
                      ↓              ↕ (tech_status loop)
                    [REVISION_REQUESTED ⇔ expert edits → PENDING → TECH_REVIEW]
                      ↓
                    TECH_APPROVED (tech_status='APPROVED')
                      ↓
                    CEO_REVIEW (unlocked only when tech_status=APPROVED)
                      ↓
                    SELECTED (ceo_status='APPROVED') / DECLINED (ceo_status='DECLINED')
```

Key scope-reduction: Mutable single row replaces `bid_versions`, `bid_revision_requests`, `price_negotiations`, `bid_conflict_overrides` tables. `tech_feedback` column replaces Surface B table. `negotiated_price_vnd` column replaces Surface C table. CEO override is implicit — CEO can APPROVE even if TECH_TEAM had REVISION_REQUESTED (no separate conflict override table needed since tech_status/ceo_status are independent columns).

MF-7: Milestone Management + DoD + Escrow (2-Layer)

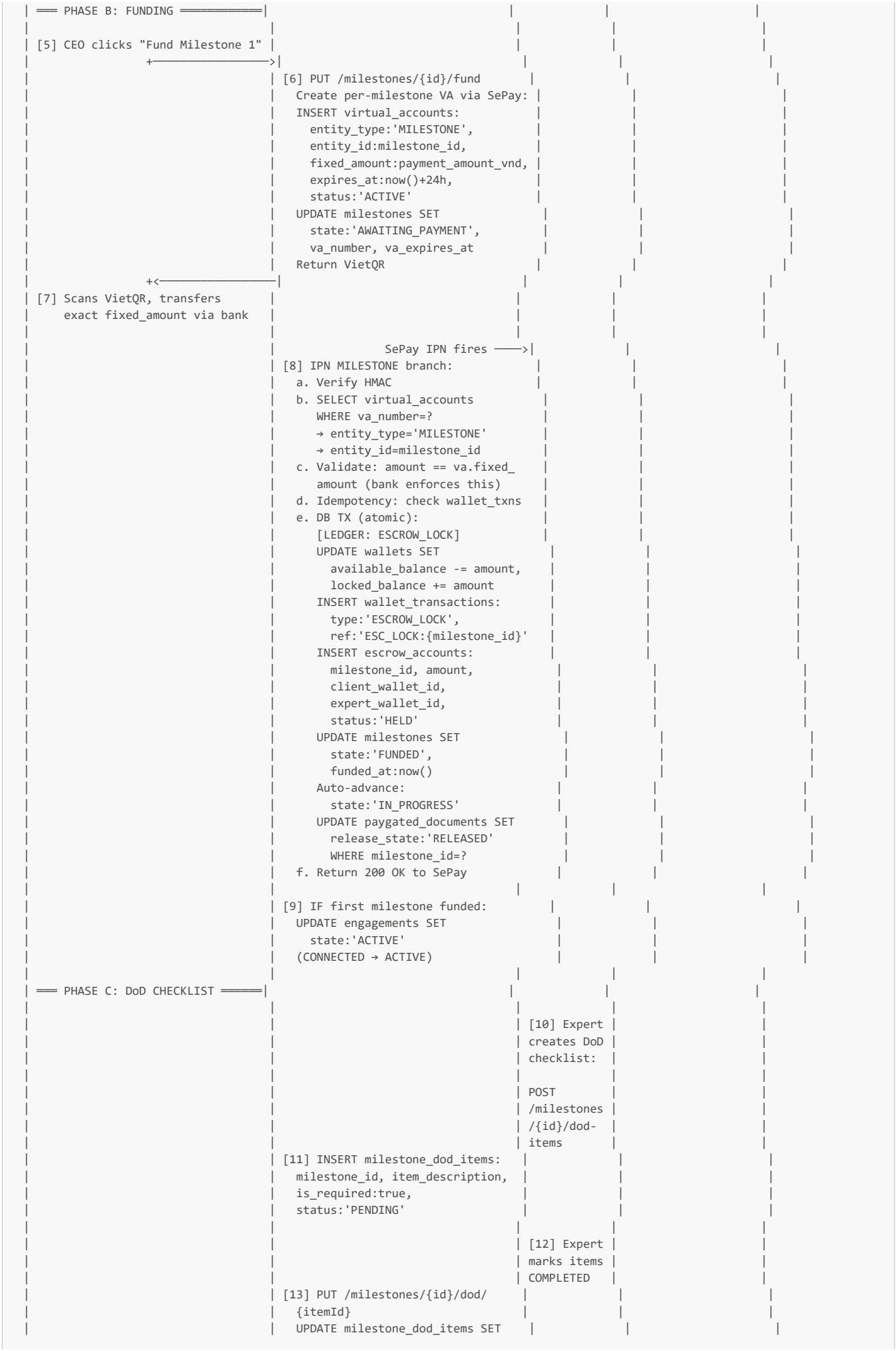
Overview

CEO funds milestones via per-milestone VA, triggering escrow lock via SePay IPN. Expert creates DoD checklist, completes items, submits deliverable. Sign-off authority verifies acceptance criteria criterion by criterion. APPROVED triggers atomic ledger release + chi hộ auto-transfer. References: §0.6 (Milestone, DoD states), §0.8 (Payment Architecture).

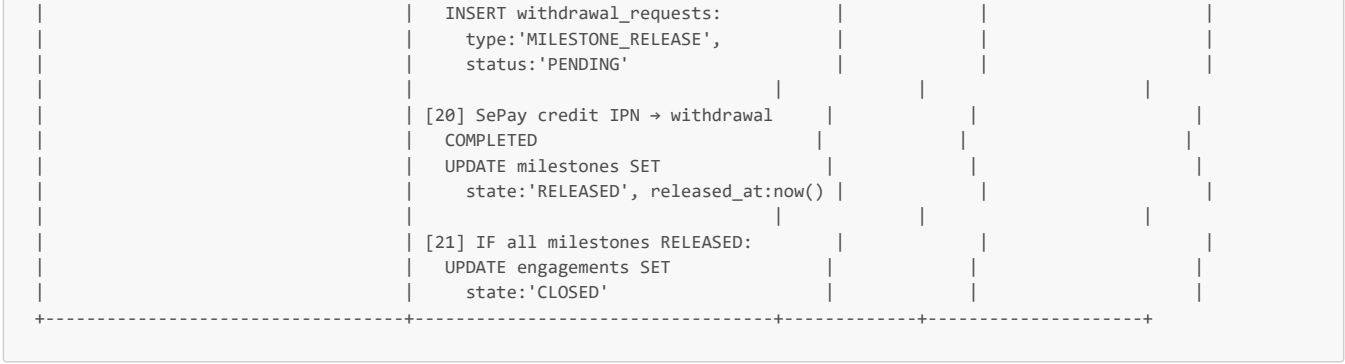
Tables touched (7): `milestones`, `acceptance_criteria`, `milestone_dod_items`, `milestone_submissions`, `paygated_documents`, `escrow_accounts`, `wallet_transactions`

Endpoints: `POST /milestones`, `POST /milestones/{id}/criteria`, `POST /milestones/{id}/dod-items`, `PUT /milestones/{id}/fund`, `PUT /milestones/{id}/dod/{itemId}`, `POST /milestones/{id}/submit`, `PUT /criteria/{id}/verify`, `POST /webhooks/sepay/ipn`

ASCII Swimlane


=== PHASE B: FUNDING ===
[5] CEO clicks "Fund Milestone 1"
[6] PUT /milestones/{id}/fund
Create per-milestone VA via SePay:
INSERT virtual_accounts:
entity_type:'MILESTONE',
entity_id:milestone_id,
fixed_amount:payment_amount_vnd,
expires_at:now()+24h,
status:'ACTIVE'
UPDATE milestones SET
state:'AWAITING_PAYMENT',
va_number, va_expires_at
Return VietQR
[7] Scans VietQR, transfers exact fixed_amount via bank
SePay IPN fires
[8] IPN MILESTONE branch:
a. Verify HMAC
b. SELECT virtual_accounts WHERE va_number=?
→ entity_type='MILESTONE'
→ entity_id=milestone_id
c. Validate: amount == va.fixed_amount (bank enforces this)
d. Idempotency: check wallet_txns
e. DB TX (atomic):
[LEDGER: ESCROW_LOCK]
UPDATE wallets SET
available_balance -= amount,
locked_balance += amount
INSERT wallet_transactions:
type:'ESCROW_LOCK',
ref:'ESC_LOCK:{milestone_id}'
INSERT escrow_accounts:
milestone_id, amount,
client_wallet_id,
expert_wallet_id,
status:'HELD'
UPDATE milestones SET
state:'FUNDED',
funded_at:now()
Auto-advance:
state:'IN_PROGRESS'
UPDATE paygated_documents SET
release_state:'RELEASED'
WHERE milestone_id=?
f. Return 200 OK to SePay
[9] IF first milestone funded:
UPDATE engagements SET
state:'ACTIVE'
(CONNECTED → ACTIVE)
=== PHASE C: DoD CHECKLIST ===
[10] Expert creates DoD checklist:
POST /milestones/{id}/dod-items
[11] INSERT milestone_dod_items:
milestone_id, item_description,
is_required:true,
status:'PENDING'
[12] Expert marks items COMPLETED
[13] PUT /milestones/{id}/dod/{itemId}
UPDATE milestone_dod_items SET


status:'COMPLETED',
completed_at:now(),
completion_note:{text}
(IF is_required=true: note req'd)
DB CHECK enforced:
NOT(is_required=true AND
status='NOT_APPLICABLE')
=== PHASE D: SUBMIT DELIVERABLE =
[14] Expert submits deliverable
[15] POST /milestones/{id}/submit
GUARD:
SELECT COUNT(*) FROM
milestone_dod_items
WHERE milestone_id=? AND
is_required=true AND
status!='COMPLETED'
IF >0 → 422 REQUIRED_DOD_INCOMPLETE
with list of unchecked items
INSERT milestone_submissions:
milestone_id, expert_id,
description, files_json,
submitted_at:now()
UPDATE milestones SET
state:'SUBMITTED',
submitted_at:now()
=== PHASE E: SIGN-OFF ==========
[16] TECH_TEAM verifies criteria criterion by criterion:
[17] PUT /criteria/{id}/verify
UPDATE acceptance_criteria SET
verified_at:now()
OR write revision_note (reject)
→ milestone state: IN_REVISION
[18] APPROVED GUARD:
SELECT COUNT(*) FROM
acceptance_criteria
WHERE milestone_id=? AND
is_required=true AND
verified_at IS NULL
IF >0 → 422 UNVERIFIED_CRITERIA
with list of unverified criteria
[19] ALL required criteria verified → APPROVED — atomic ledger release:
DB TX:
[LEDGER: ESCROW_RELEASE]
UPDATE wallets (client) SET
locked_balance -= amount
[LEDGER: PLATFORM_FEE]
fee = amount * platform_fee_pct
(READ from platform_settings, NOT hardcoded)
UPDATE wallets (platform) SET
available_balance += fee
[LEDGER: CREDIT_EXPERT]
net = amount - fee
UPDATE wallets (expert) SET
available_balance += net
INSERT wallet_transactions (3 rows):
ESCROW_RELEASE, PLATFORM_FEE,
(expert credit via ESCROW_RELEASE)
UPDATE escrow_accounts SET
status:'RELEASED', released_at:now()
UPDATE milestones SET
state:'APPROVED', approved_at:now()
COMMIT
→ chi hộ API called (async)
[API] POST SePay chi hộ:
{amount:net, bank_account_xid,
reference:"WD-{withdrawal_id}"}



Step Detail Table

Step	Actor	Action	Tables (CRUD)	State Change	Endpoint
1-2	CEO	Create milestone	`milestones` (C)	state: DEFINED	`POST /milestones`
3-4	CEO→FastAPI	Define criteria + LLM quality gate	`acceptance_criteria` (C), `platform_decisions` (C)	verified_at: NULL	`POST /milestones/{id}/criteria`
5-6	CEO	Fund milestone → VA created	`virtual_accounts` (C), `milestones` (U)	state: DEFINED→AWAITING_PAYMENT	`PUT /milestones/{id}/fund`
7-8	CEO→SePay→NestJS	Scan QR → IPN → escrow lock	`virtual_accounts` (R), `wallets` (U), `wallet_transactions` (C), `escrow_accounts` (C), `milestones` (U), `paygated_documents` (U)	state: AWAITING_PAYMENT→FUNDED→IN_PROGRESS; escrow: HELD	`POST /webhooks/sepay/ipn`
9	NestJS	First milestone → engagement ACTIVE	`engagements` (U)	state: CONNECTED→ACTIVE	—
10-11	Expert	Create DoD items	`milestone_dod_items` (C)	status: PENDING	`POST /milestones/{id}/dod-items`
12-13	Expert	Complete DoD items	`milestone_dod_items` (U)	status: PENDING→COMPLETED	`PUT /milestones/{id}/dod/{itemId}`
14-15	Expert	Submit deliverable (DoD guard)	`milestone_submissions` (C), `milestones` (U)	state: IN_PROGRESS→SUBMITTED	`POST /milestones/{id}/submit`
16-17	TECH_TEAM	Verify criteria	`acceptance_criteria` (U)	verified_at set (or revision_note written)	`PUT /criteria/{id}/verify`
18-19	System	All required criteria verified → APPROVED + ledger release	`wallets` (U ×3), `wallet_transactions` (C ×3), `escrow_accounts` (U), `milestones` (U), `withdrawal_requests` (C)	state: SUBMITTED→APPROVED; escrow: HELD→RELEASED	—
20	SePay→NestJS	Chi hộ credit IPN	`withdrawal_requests` (U), `milestones` (U)	state: APPROVED→RELEASED; withdrawal: COMPLETED	`POST /webhooks/sepay/ipn`
21	NestJS	All milestones RELEASED → engagement CLOSED	`engagements` (U)	state: ACTIVE→CLOSED	—

Milestone State Machine (§0.6 — Scope-Reduced)

```
DEFINED → AWAITING_PAYMENT → FUNDED → IN_PROGRESS → SUBMITTED → APPROVED → RELEASED
                                                      ↕            ↑
                                                  IN_REVISION ─────┘
                                                      ↓
                                                  DISPUTED
```

MF-8: Dispute Resolution (2-Layer)

Overview

Dispute filed against a specific acceptance criterion. Layer 1: LLM evaluates criterion vs. deliverable. If confidence ≥ 0.80 → AUTO_RESOLVED. If < 0.80 → MANUAL_REVIEW → admin resolves via dashboard button. References: §0.6 (Dispute states), §0.8 (Ledger operations).

Tables touched (5): `disputes`, `escrow_accounts`, `wallet_transactions`, `wallets`, `platform_decisions`

Endpoints: `POST /disputes`, `PUT /admin/disputes/{id}/resolve`

ASCII Swimlane

Dispute State Machine (2-Layer — §0.6)

```
PENDING → LAYER_1_EVAL
          → AUTO_RESOLVED (confidence ≥ 0.80) [LEDGER]
          → MANUAL_REVIEW (confidence < 0.80)
             → RESOLVED (admin button) [LEDGER]
```

Key scope-reduction: No Layer 2 mutual agreement (48h cooling window) or Layer 3 automatic 50/50 split. Admin resolves Layer 2 manually with 3 options. No `dispute_resolution_reports` table.

Group 3 — Path B: Service Marketplace

MF-9: Expert Service Publishing (AI Generator)

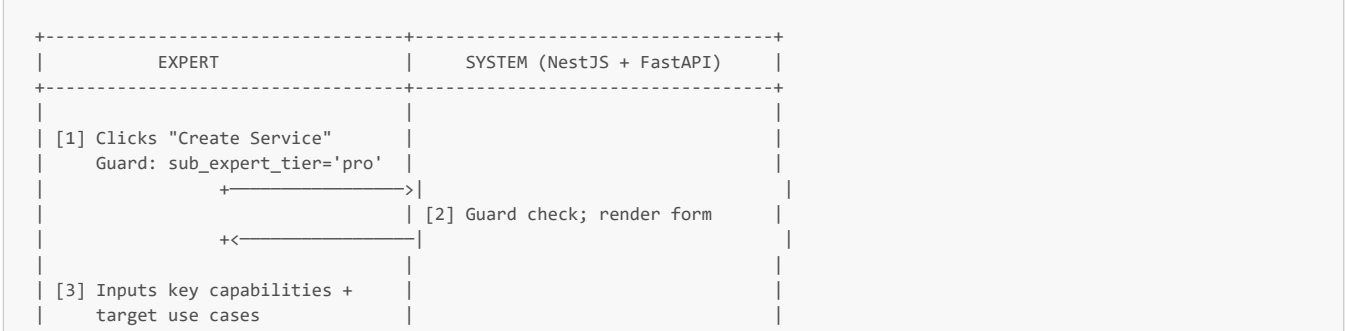
Overview

Expert (Pro) uses AI Service Generator to create a service listing. Publishes to marketplace. References: §0.9 (Expert Pro required for AI Service Generator).

Tables touched (2): `services`, `expert_profiles`

Endpoints: `POST /services/generate`, `POST /services`, `PUT /services/{id}`

ASCII Swimlane



MF-10: Client Buys Service / Tech Discovery

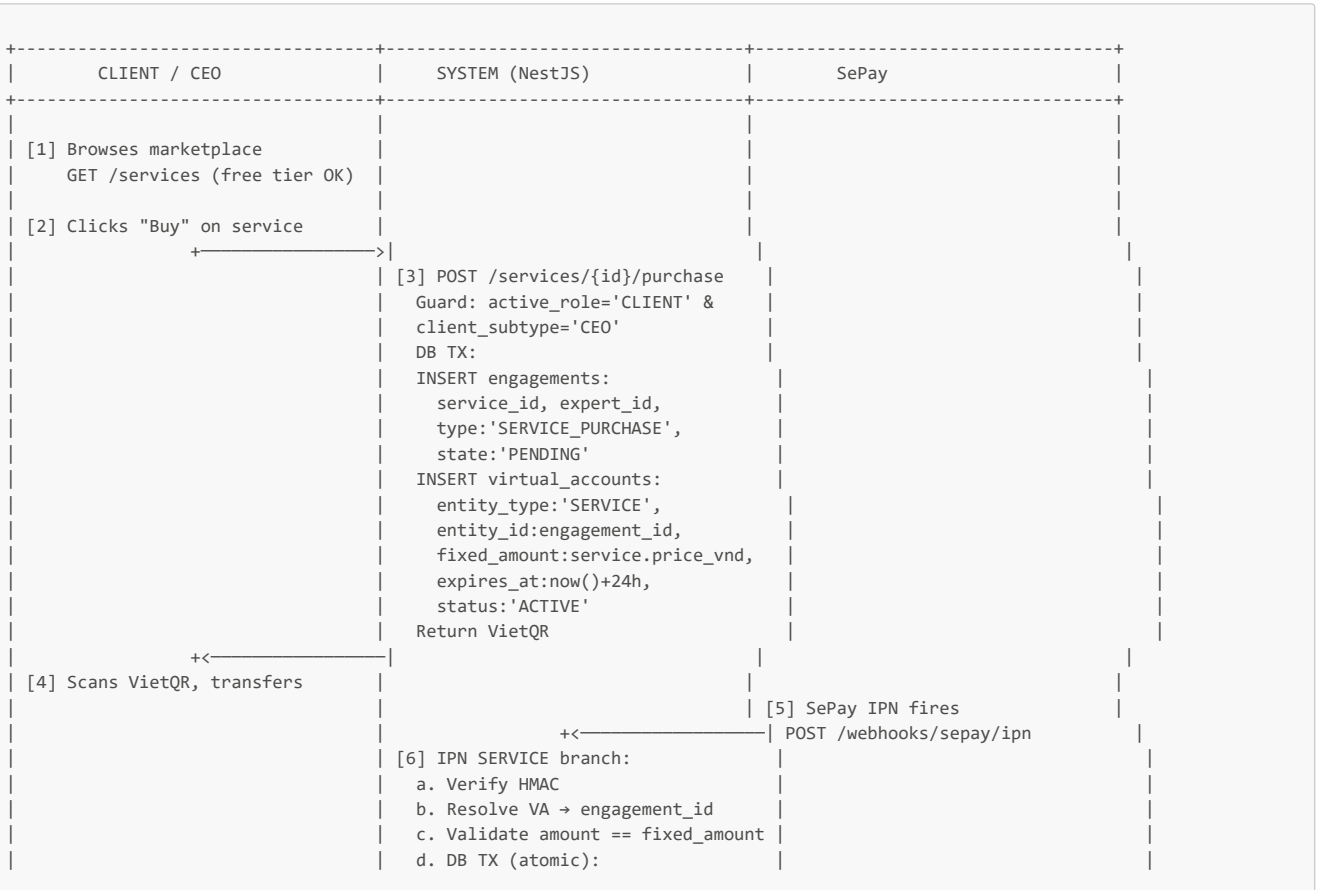
Overview

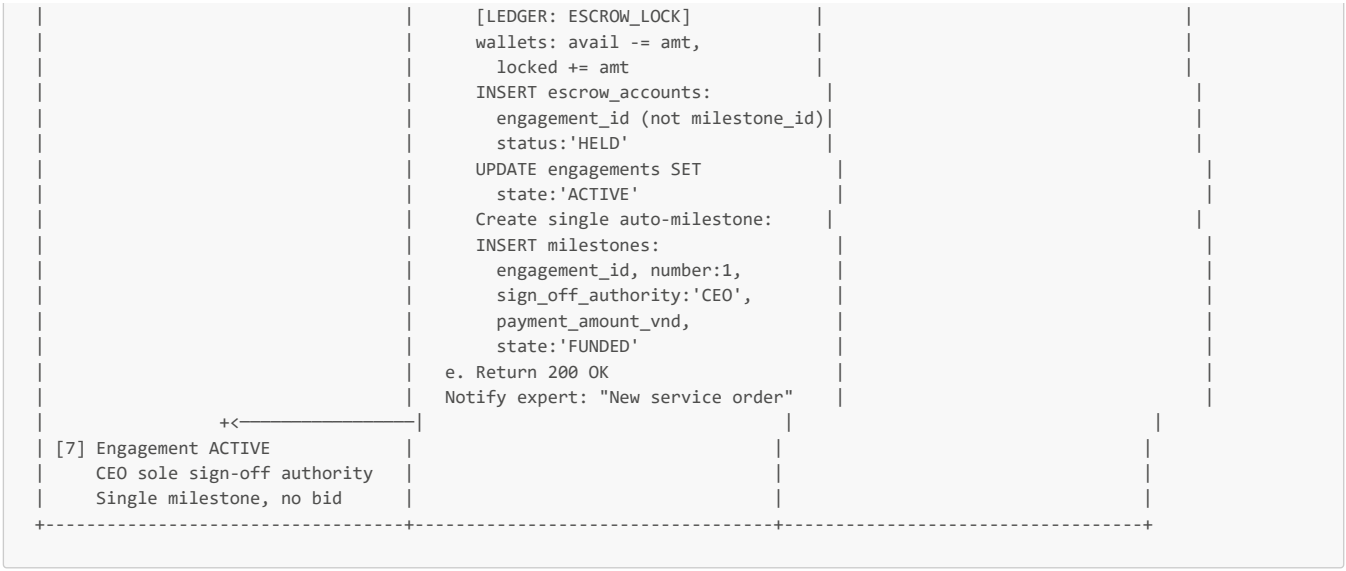
CEO browses marketplace, purchases a service via per-order VA payment. Creates SERVICE_PURCHASE or TECH_DISCOVERY engagement directly in ACTIVE state (no bid, no elicitation). References: §0.6 (Engagement types), §0.8 (Payment Architecture).

Tables touched (5): `services`, `engagements`, `virtual_accounts`, `escrow_accounts`, `wallet_transactions`

Endpoints: `GET /services`, `POST /services/{id}/purchase`, `POST /webhooks/sepay/ipn`

ASCII Swimlane





Key structural difference from Path A: No elicitation, no matching, no bid, no TECH_TEAM involvement. Single fixed-price milestone with CEO-only sign-off. `engagements.project_id = NULL`, `engagements.service_id NOT NULL`. Enforced by table-level CHECK constraint.

Group 4 — Financial Flows

MF-11: Wallet Top-Up

Overview

User scans permanent WALLET_TOPUP VA VietQR. SePay IPN credits wallet. Pure internal ledger — no external API calls.

Tables touched (3): `virtual_accounts`, `wallets`, `wallet_transactions`

Endpoints: `POST /wallet/topup-qr`, `POST /webhooks/sepay/ipn`

ASCII Swimlane

```
| [6] Balance updated              |                                  |                                  |
+----------------------------------+----------------------------------+----------------------------------+
```

Idempotency guarantee: `wallet_transactions` has `UNIQUE INDEX wallet_tx_idempotency ON (wallet_id, reference_id) WHERE reference_id IS NOT NULL`. SePay IPN retries cannot double-credit.

MF-12: Expert Withdrawal (Chi Hộ)

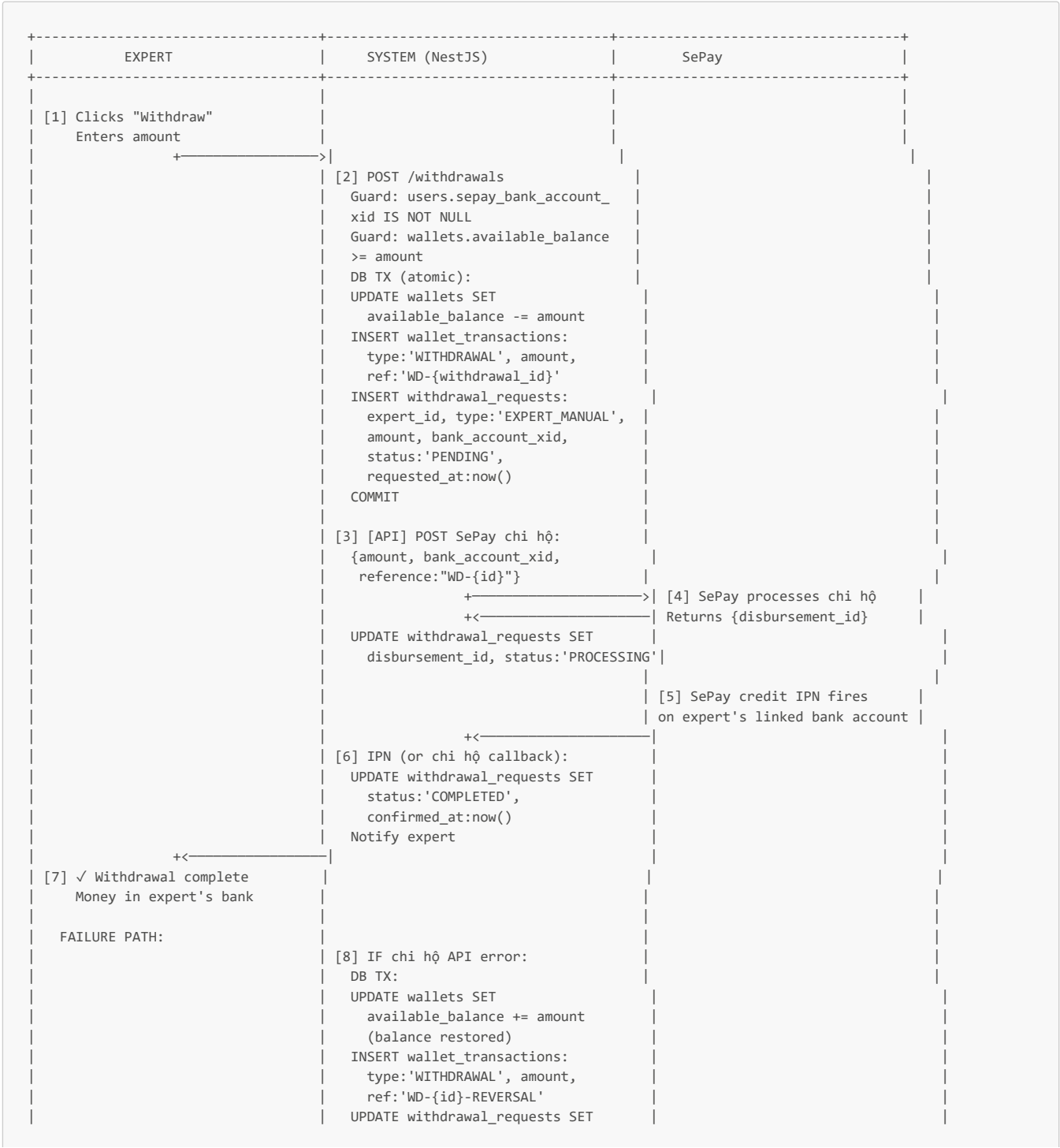
Overview

Expert requests cash-out. System debits wallet atomically, calls SePay chi hộ API. Credit IPN confirms completion. Zero admin involvement.

Tables touched (3): `wallets`, `wallet_transactions`, `withdrawal_requests`

Endpoints: `POST /withdrawals`, `POST /webhooks/sepay/ipn`

ASCII Swimlane



```
|                                  |     status:'FAILED'                  |                                  |
|                                  |   Notify expert with error details   |                                  |
+----------------------------------+--------------------------------------+----------------------------------+
```

MF-13: Subscription Purchase from Wallet

Overview

Pure internal wallet deduction. No VA, no IPN. Atomic transaction debits wallet and updates user subscription columns.

Tables touched (3): `users`, `wallets`, `wallet_transactions`

Endpoints: `POST /subscriptions/activate`

ASCII Swimlane

Key scope-reduction: No `user_subscriptions` table. Subscription state stored as `users.subscription_client_tier`, `users.sub_client_expires_at`, `users.subscription_expert_tier`, `users.sub_expert_expires_at`. The `wallet_transactions` reference_id pattern `SUB-{user_id}:{role_type}` provides the audit trail that the cut table would have provided.

Group 5 — Platform-Specific Flows

MF-14: Pay-Gated Document Release (IPN-Triggered)

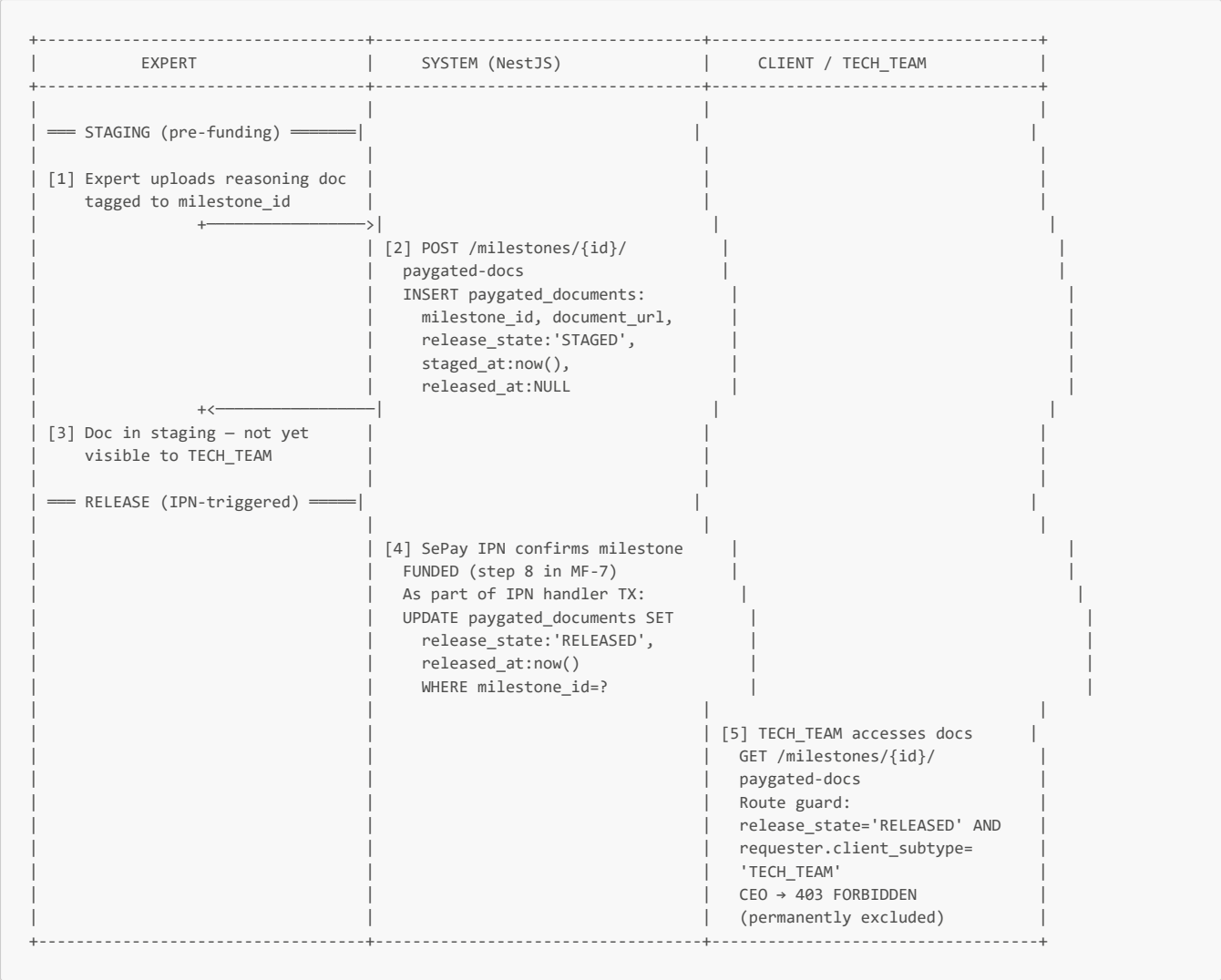
Overview

Core F5 IP deadlock resolution mechanism. Expert stages reasoning documents tagged to a milestone. When SePay IPN confirms milestone FUNDED, documents auto-release to TECH_TEAM. CEO permanently excluded. References: §0.7 (RBAC — TECH_TEAM-only access).

Tables touched (2): `paygated_documents`, `milestones`

Endpoints: `POST /milestones/{id}/paygated-docs`, `GET /milestones/{id}/paygated-docs`

ASCII Swimlane



MF-15: Dual-Role Account Switch

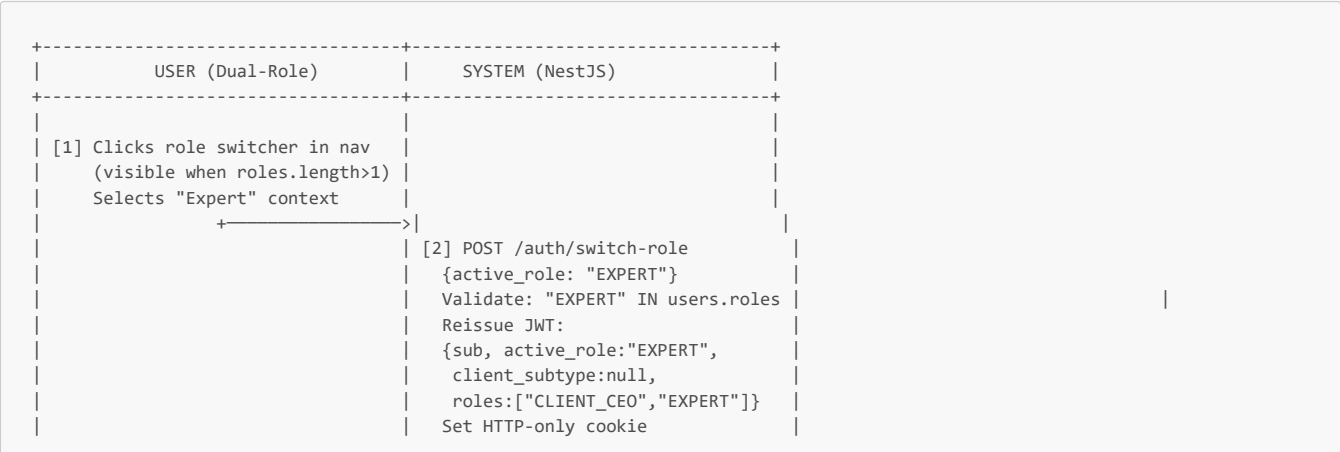
Overview

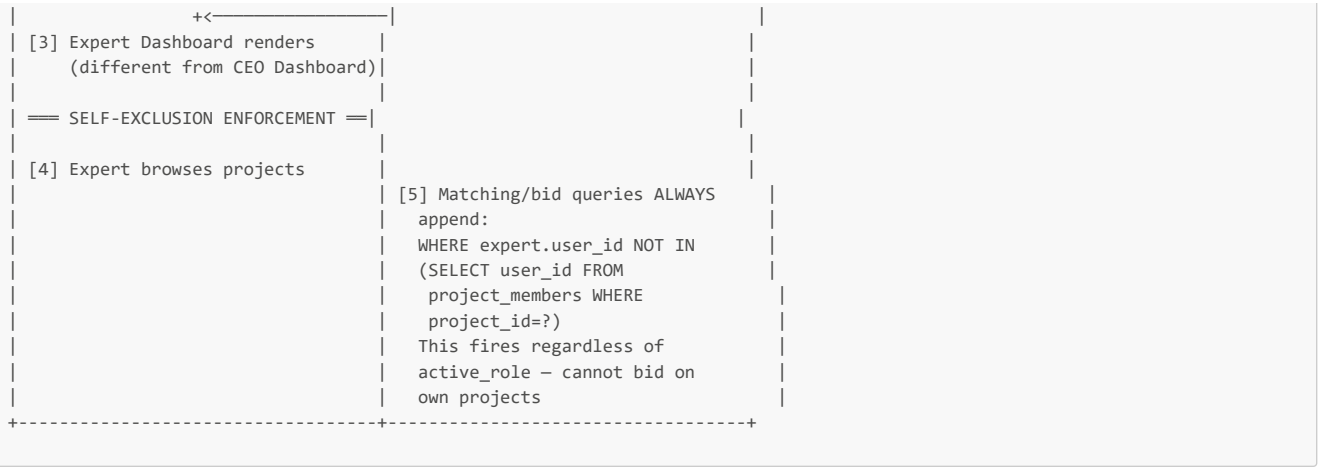
Single account holds `roles: ["CLIENT_CEO", "EXPERT"]`. Role switcher reissues JWT with new `active_role`. Self-exclusion always applies. References: §0.7 (Dual-role, self-exclusion).

Tables touched (1): `users`

Endpoints: `POST /auth/switch-role`

ASCII Swimlane





MF-16: Expert Portfolio Tier 2 Auto-Upgrade

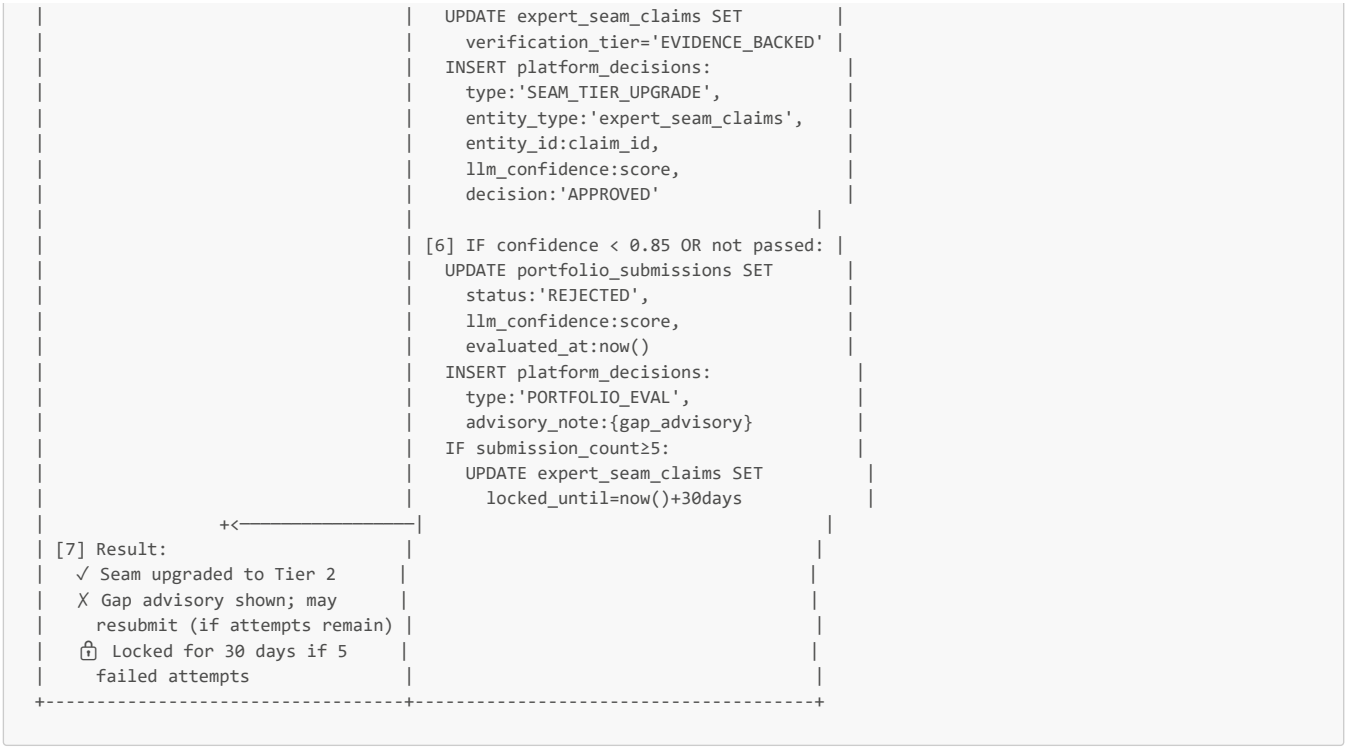
Overview

Dedicated flow for the LLM portfolio verification pipeline. Expert submits structured decision-point writeup. FastAPI evaluates. Auto-upgrades seam claim to EVIDENCE_BACKED if confidence ≥ 0.85. 5-attempt / 30-day lockout throttle. References: §0.4 (2-Tier Verification).

Tables touched (3): `portfolio_submissions`, `expert_seam_claims`, `platform_decisions`

Endpoints: `POST /portfolio-submissions`, `GET /portfolio-submissions/{id}/eval-result`

ASCII Swimlane



Group 6 — Admin Flows

MF-17: Platform Integrity Monitor & Analytics

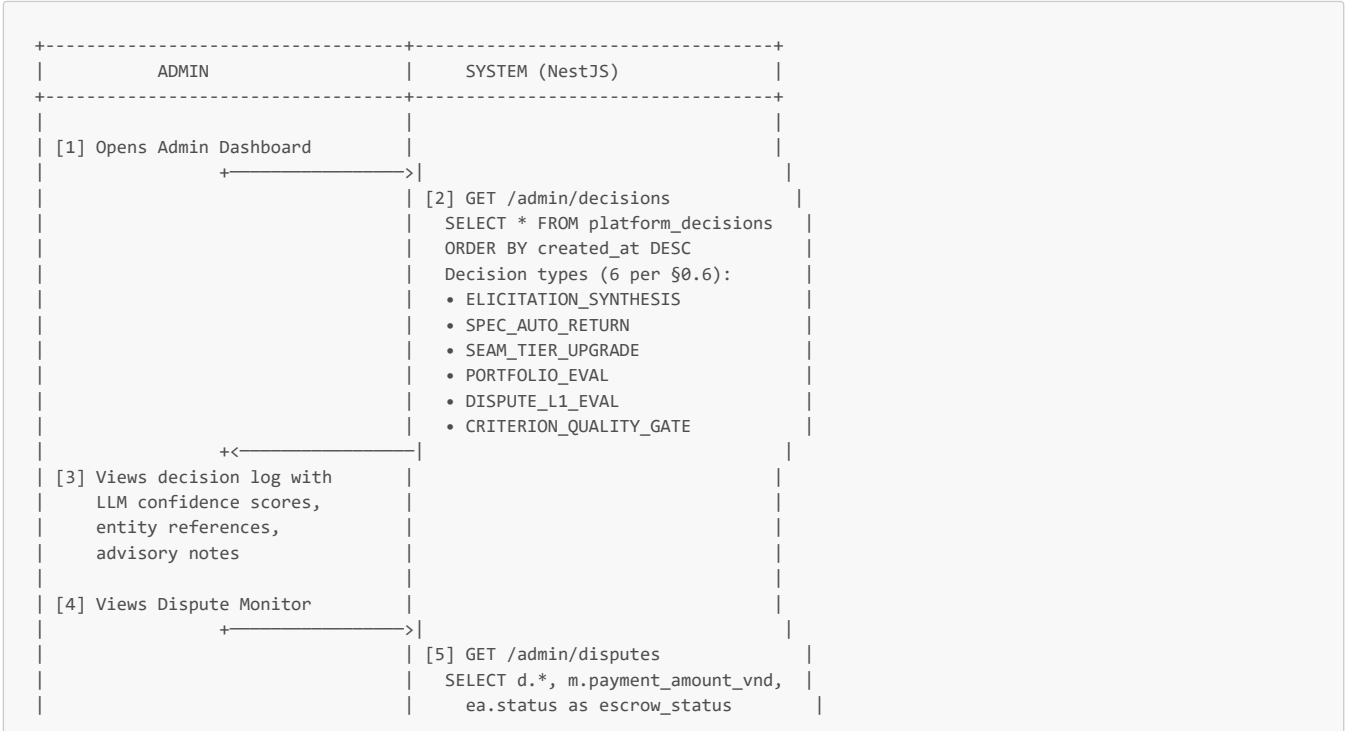
Overview

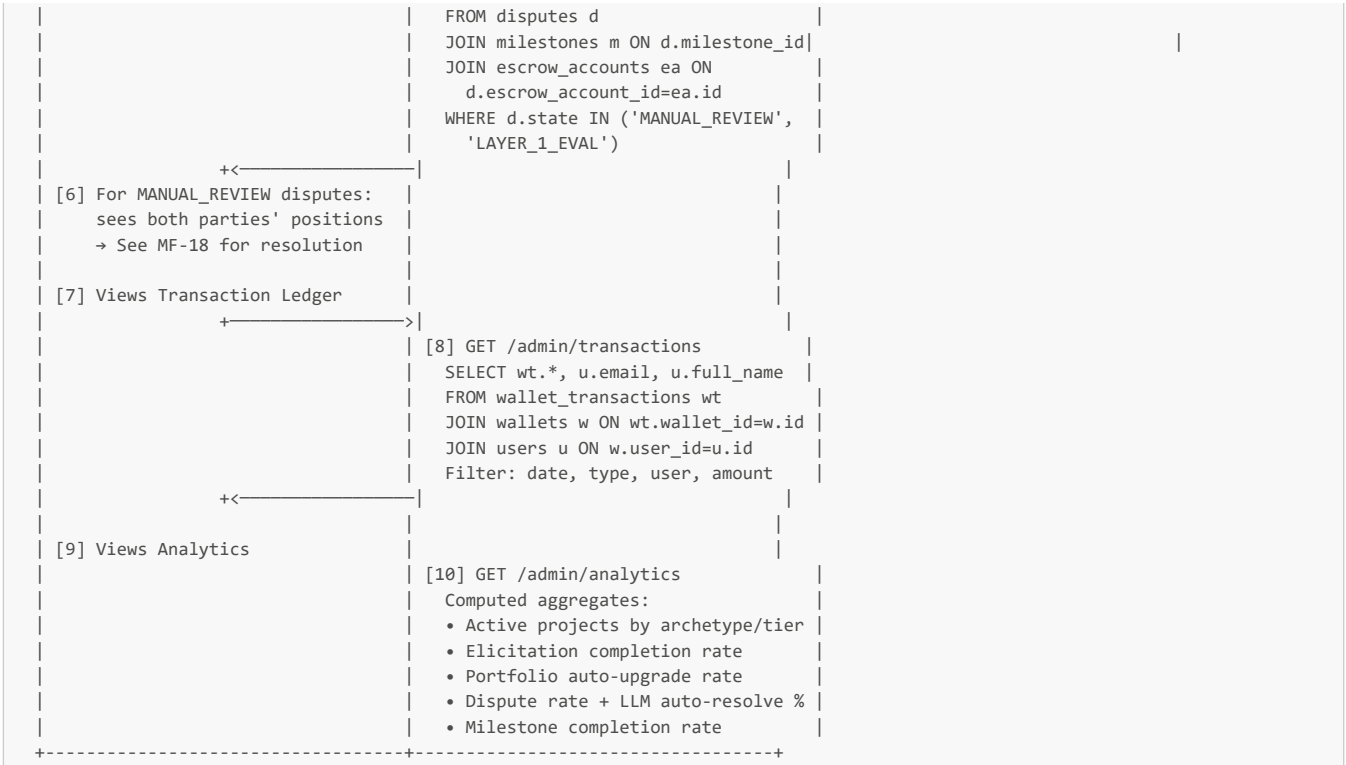
Read-only monitoring dashboard. Displays `platform_decisions` log, auto-upgrade/return history, dispute log, transaction ledger, analytics. References: §0.7 (ADMIN — read-only monitoring).

Tables touched (1 — read only): `platform_decisions` (plus read-only access to `wallet_transactions`, `disputes`, `escrow_accounts`, `users`, `withdrawal_requests`)

Endpoints: `GET /admin/decisions`, `GET /admin/disputes`, `GET /admin/transactions`, `GET /admin/analytics`

ASCII Swimlane





MF-18: Manual Dispute Resolution & Account Management

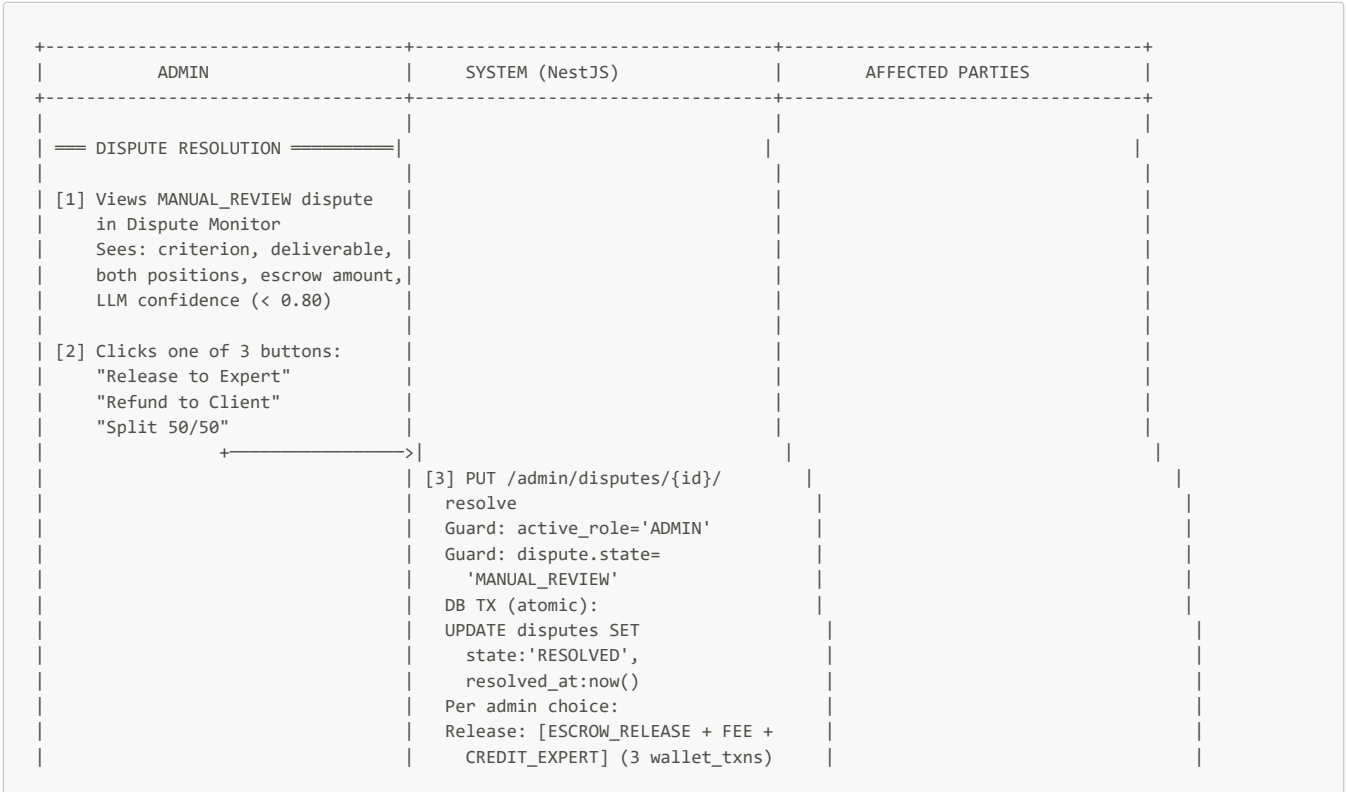
Overview

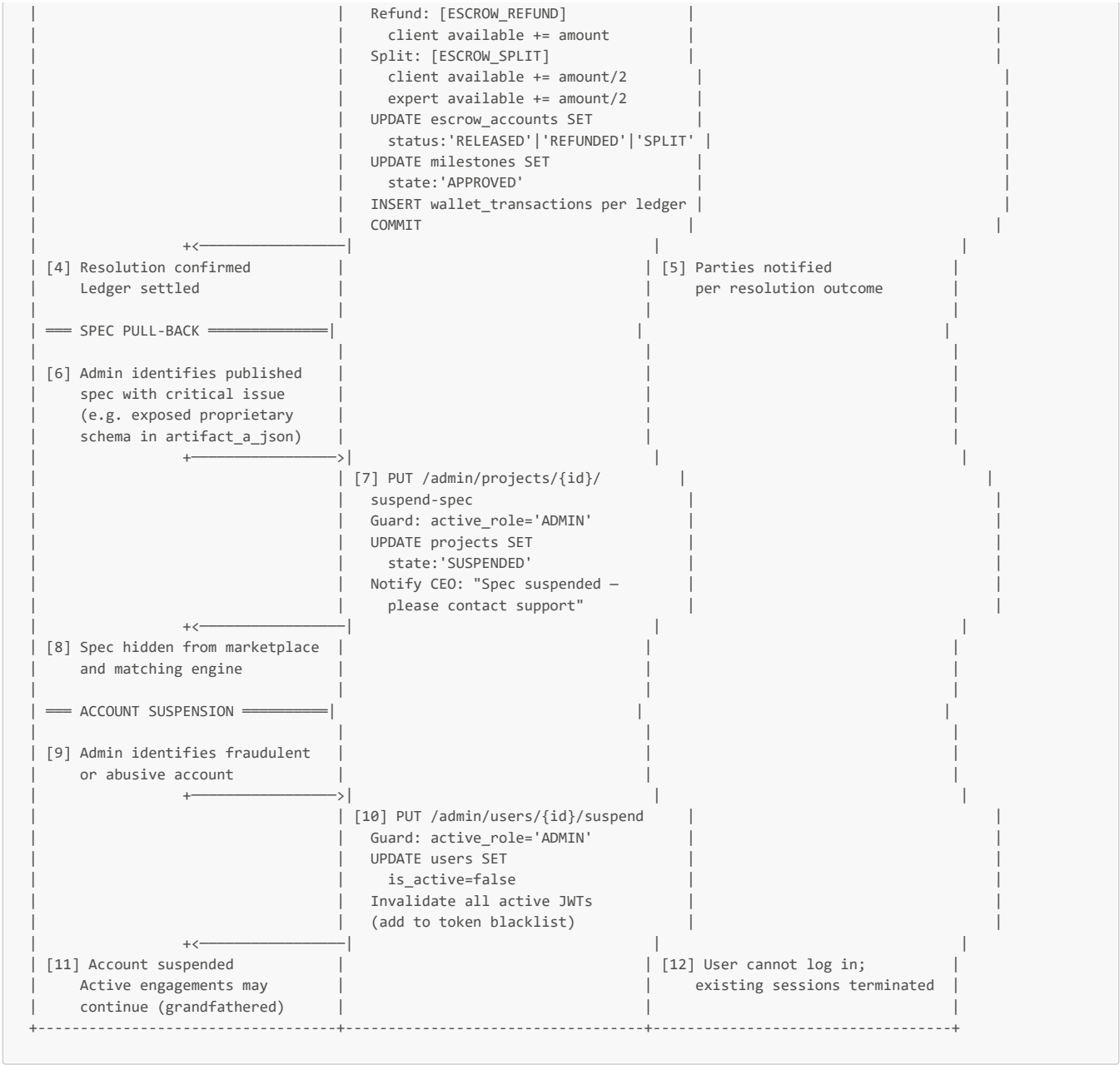
Admin's only write actions: (1) resolve MANUAL_REVIEW disputes via dashboard button, (2) emergency spec pull-back, (3) account suspension. All three trigger atomic ledger or state operations. References: §0.7 (ADMIN — emergency write actions).

Tables touched (4): `disputes`, `escrow_accounts`, `wallets`, `wallet_transactions`, `projects`, `users`

Endpoints: `PUT /admin/disputes/{id}/resolve`, `PUT /admin/projects/{id}/suspend-spec`, `PUT /admin/users/{id}/suspend`

ASCII Swimlane





Cross-Table Ledger Operations Summary

Every financial action in the platform resolves to one of these atomic wallet transaction patterns:

Transaction Type	Wallet Debit	Wallet Credit	Trigger	Flow
`TOP_UP`	—	`available_balance += amount`	SePay IPN (WALLET_TOPUP VA)	MF-11
`SUBSCRIPTION`	`available_balance -= price`	—	User activates Pro	MF-13
`ESCROW_LOCK`	`available_balance -= amount`	`locked_balance += amount`	SePay IPN (MILESTONE/SERVICE VA)	MF-7, MF-10
`ESCROW_RELEASE`	`locked_balance -= gross`	`expert.available_balance += net`	All required criteria verified	MF-7
`PLATFORM_FEE`	—	`platform.available_balance += fee`	Deducted on release	MF-7
`ESCROW_REFUND`	`locked_balance -= amount`	`client.available_balance += amount`	Dispute: client wins	MF-8, MF-18
`ESCROW_SPLIT`	`locked_balance -= amount`	Both `+= amount/2`	Admin 50/50 split	MF-18
`WITHDRAWAL`	`expert.available_balance -= amount`	—	Expert cash-out request	MF-12

Every row in `wallet_transactions` is immutable. The `UNIQUE INDEX wallet_tx_idempotency ON (wallet_id, reference_id) WHERE reference_id IS NOT NULL` guarantees SePay IPN retries cannot double-credit.

Platform fee is never hardcoded: `SELECT platform_fee_pct FROM platform_settings LIMIT 1` is read at transaction time. Default 0.05 (5%). Can be changed via DB without a code deploy.

Appendix: 28-Table Coverage Matrix

> **Every table in the schema is exercised by at least one main flow.** This matrix proves full CRUD coverage.

#	Table	Created By	Read By	Updated By	Primary Flows
1	`users`	MF-1, MF-2, MF-3	MF-1, MF-2, MF-5, MF-13, MF-15	MF-1 (sub tier), MF-2 (bank link), MF-13 (sub tier)	All
2	`client_profiles`	MF-1	—	—	MF-1
3	`expert_profiles`	MF-2	MF-5, MF-9	MF-2 (stack tags, archetype)	MF-2, MF-5
4	`tech_team_profiles`	MF-3	MF-3		